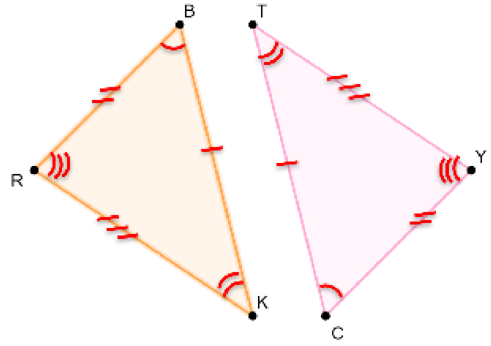
Unit 5 Overview: Triangle Congruence

Unit Narrative

Students have been introduced to congruence and similarity in previous units, even identifying congruent triangles in Unit 1 Lesson 7. While students have engaged with the basic structure and vocabulary associated with triangles (angles, side lengths, base, legs) and congruence, Unit 5 extends these working definitions to include postulates and theorems used to prove triangle congruence.



Section 1 lays the foundation for the general theme of the unit – triangle congruence. In Lesson 1, students will learn the first of several postulates and theorems, Side-Side-Side Congruence Postulate before transitioning to more postulates and theorems in Lessons 2-4. In Lesson 4, students begin to relate what they have learned about congruence to the concept of rigid motions that they have learned in prior units. Additionally, students will explore the concept of corresponding parts of congruent triangles to write congruence statements.

Section 2 shifts from definitions of different postulates and theorems to drawing conclusions and proving triangles congruent using those postulates and theorems. Lesson 5 is an opportunity to activate prior knowledge as well as identify and address common misconceptions. Students then progress into Lesson 6, where they begin formalizing what they did in the previous lesson to complete proofs in various styles such as flowchart, paragraph, and two-column. In this lesson, students work with what they learned about SSS, SAS, ASA, AAS, or HL congruence to complete proofs. In Lesson 7, students will continue their work with proving triangles congruent, but will work further with the concept of Corresponding Parts of Congruent Triangles are Congruent, or CPCTC, as a part of the formal proof.

Unit At-A-Glance

Section 1: Triangle Congruence	Benchmarks Addressed	Lesson 1	Exploring Side-Side-Side Triangle Congruence
	MA.912.GR.1.2 **MA.912.GR.1.6** **MA.912.GR.2.1** **MA.912.GR.2.3** **MA.912.GR.2.6** **MA.912.GR.5.1**	Lesson 2	Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence
		Lesson 3	Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence
		Lesson 4	Exploring Corresponding Parts of Congruent Triangles Are Congruent
Section 2: Proving Triangles Congruent	Benchmarks Addressed	Lesson 5	Drawing Conclusions from Given Information
	MA.912.GR.1.2 **MA.912.GR.1.6** **MA.912.GR.2.7** *(Honors only)* **MA.912.LT.4.10**	Lesson 6	Proving Triangles Congruent
		Lesson H1	Definition of Congruence in Terms of Rigid Transformations
		Lesson 7	Corresponding Parts of Congruent Triangles Are Congruent
		Lesson 8	The Proof Is in the Bag!
Section 3: Using Congruence of Two-Dimensional Figures	Benchmarks Addressed	Lesson 9	Congruence in Two-Dimensional Figures – Part 1
	MA.912.GR.1.6	Lesson 10	Congruence in Two-Dimensional Figures – Part 2

Differentiation

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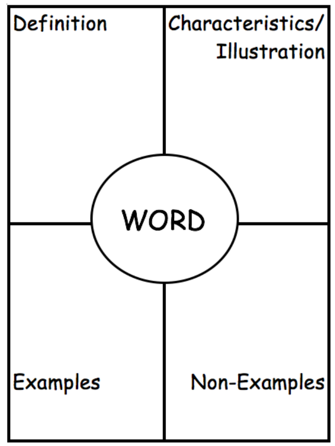
Engagement *The "WHY" of learning*	• Triangle Congruence in the Real World: Students engage with a variety of examples to make meaning of triangles and congruence in the real world. For example, the Warm-Ups in Lessons 1 (pictured here) and 2 offer unique entry points to conversations on triangles in STEM career fields to infuse meaning into the unit. Consider highlighting one of the connections from these lessons that your students particularly identify with in order to more effectively engage students in the meaning and purpose of triangle congruence. 
Representation *The "WHAT" of learning*	• Postulates and Theorems: The vocabulary of postulates and theorems were introduced in Unit 1. Triangle congruence theorems and postulates can vary as to which is a postulate and which is a theorem. As the focus of this course is not proving a postulate from a theorem, whether a student uses postulate or theorem in their reasons does not matter. Employ vocabulary retention strategies such as anchor charts, 3x5 cards, student notebooks, flip charts, Frayer Models, etc. to help students understand each theorem or postulate and how it is used to determine triangle congruence.

Action & Expression
The "HOW" of learning
	• Language Acquisition Strategies: Sentence Frames and Scaffolding
	Unit 5 requires that students demonstrate their learning by both engaging with images as well as providing verbal or written descriptions as they construct proofs and make meaningful connections between what they see, what they know, and how to communicate. For support in writing the statements and reasons in a proof, consider creating a foldable or list resource of definitions, postulates, and theorems seen in the proofs throughout the unit. Include typical "next statements" after a definition, postulate, or theorem is used in a proof.
	For example, After stating two lines are perpendicular, a typical next statement is
	  • "…the angles formed are right angles by definition of perpendicular lines."
	After stating an angle is a right angle, a typical next statement is
	  • "…a triangle is a right triangle by definition of a right triangle."
	  • "…the measure of an angle is $90°$by definition of a right angle."

ELL Information

Students will be identifying and interpreting key features of graph of functions. This unit provides an opportunity to review some important vocabulary. Have students create a Frayer Model for the words in this topic:

- A Frayer Model (similar to the four-square graphic organizer) is helpful because it allows students to see terms in multiple ways. For this section, students will be introduced to many words that set the foundation for functions.
- Provide the definition, but have students work in pairs to illustrate the word. (When possible, or else list characteristics, list examples, and list non-examples).
- For examples and nonexamples, encourage students to consider this a work in progress, and to add throughout the section as they are presented.
- Consider creating a Frayer Model for each term from the Glossary by Section lists (below).



For English Language Learners, consider using the Spanish translation for each of the postulates and theorems in this unit. The visuals for each one should be consistent with those provided in the English version of the same postulate and theorem so that students can connect to the content in their native language but understand the general concept of triangle congruence using the correct markings and congruency statements.

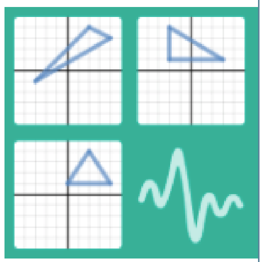
All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support” below:

- Congruent
- Rigid Transformation (Motion)

Additional Differentiated Support

Scaffolding Resources

Digital Learning Tools



In addition to digital scaffolding like Study Expert videos and other tools in the Student Area, consider other digital resources that provide additional entry points and means for making sense of triangles, congruence, and similarity. For example, Desmos features “Polygraph: Triangles” (pictured here) that offers students more opportunities to engage in basic definitions of features of triangles, particularly when comparing different triangles. Although this activity does not intentionally target congruence, the activity can be used as an engaging, low-floor discussion for building fluency in basic features and functions of triangles and congruence.

On-Ramp

Use the videos and practice problems in the On-Ramp to Geometry personalized learning tool to review the following topics for this unit. Videos are available in English and Spanish.

- Perimeter and Area on the Coordinate Plane
- Identifying Transformations

Section Overview

<table>
<tr><th colspan="3">Section 1: Triangle Congruence (Lessons 1-4)</th></tr>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="3">

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

- *Clarification 1: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.*
- *Clarification 2: Instruction focuses on helping a student chose a method they can use reliably.*

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

- *Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.*

MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates.

- *Clarification 1: Instruction includes the connection of transformations to functions that take points in the plane as inputs and give other points in the plane as outputs.*
- *Clarification 2: Transformations include translations, dilations, rotations, and reflections described using words or using coordinates.*
- *Clarification 3: Within the Geometry course, rotations are limited to $90°$, $180°$ and $270°$ counterclockwise or clockwise about the center of rotation, and the centers of rotations and dilations are limited to the origin or a point on the figure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

</td><td>Lesson 1</td><td>

- I can use constructions and transformations to determine if SSS congruence can be used to show that two triangles are congruent.

</td></tr>
<tr><td>Lesson 2</td><td>

- I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent.
- I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent.
- I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.

</td></tr>
<tr><td>Lesson 3</td><td>

- I can use constructions to determine if AAS congruence can be used to show that two triangles are congruent.
- I can determine if HL congruence can be used to show that two triangles are congruent using deduction.

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Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.
Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to $90°$, $180°$ and $270°$ counterclockwise or clockwise about the center of rotation.
Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.

MA.912.GR.5.1 Construct a copy of a segment or an angle.

Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding or dynamic geometric software.

</td>
<td>Lesson 4</td>
<td>

I can relate the definition of congruence in terms of rigid motions to corresponding parts of congruent triangles.
I can use corresponding parts of congruent triangles to write congruence statements.

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</tr>
</table>

Section 2: Proving Triangles Congruent (Lessons 5-8)

<table>
<tr>
<th>Benchmarks Addressed</th>
<th colspan="2">Learning Targets</th>
</tr>
<tr>
<td rowspan="2">
MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg.

Clarification 1: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.
Clarification 2: Instruction focuses on helping a student chose a method they can use reliably.

</td>
<td>Lesson 5</td>
<td>

I can draw and mark figures using given information.
I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from given information.

</td>
</tr>
<tr>
<td>Lesson 6</td>
<td>

I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.

</td>
</tr>
</table>

MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. • *Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.* **MA.912.GR.2.7** (Honors only) Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations. **MA.912.LT.4.10** Judge the validity of arguments and give counterexamples to disprove statements. • *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs.*	Lesson H1	• I can use the definition of congruence to justify that SSS congruence, SAS congruence, ASA congruence, and AAS congruence are sufficient to show that two triangles are congruent.
	Lesson 7	• I understand how to use corresponding parts of congruent triangles in a proof. • I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.
	Lesson 8	• I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence. • I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

Section 3: Using Congruence of Two-Dimensional Figures (Lessons 9-10)

Benchmarks Addressed	Learning Targets	
MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. • *Clarification 1: Instruction includes demonstrating that two-dimensional figures are congruent or similar based on given information.*	Lesson 9	• I can solve mathematical problems involving congruence.
	Lesson 10	• I can solve real-world problems involving congruence.

Vertical Alignment

Grade 8 Unit 9
Transformations

Geometry 1 Unit 5
Triangle Congruence

Prior Course(s)	Course Level	Future Course(s)
MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles. **MA.8.GR.2.1** Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship. **MA.8.GR.2.2** Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.	**MA.912.GR.1.2** Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle and Hypotenuse-Leg. **MA.912.GR.1.6** Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. **MA.912.GR.2.1** Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. **MA.912.GR.2.3** Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure. **MA.912.GR.2.6** Apply rigid transformations to map one figure onto another to justify that the two figures are congruent. **MA.912.GR.2.7** *(Honors only)* Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations. **MA.912.GR.5.1** Construct a copy of a segment or an angle. **MA.912.LT.4.10** Judge the validity of arguments and give counterexamples to disprove statements.	N/A

5.1 Exploring Side-Side-Side Triangle Congruence

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.2.1 Given a preimage and image, describe the transformation and represent the transformation algebraically using coordinates. MA.912.GR.5.1 Construct a copy of a segment or an angle.	Learning Target	• I can use constructions and transformations to determine if SSS congruence can be used to show that two triangles are congruent.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions

- How can I use transformations to determine if SSS congruence can be used to show that two triangles are congruent?
- How can I use constructions to determine if SSS congruence can be used to show that two triangles are congruent?

Lesson Narrative

In this lesson, students will explore the side lengths of triangles to discover the Side-Side-Side (SSS) congruence postulate. They will use this information to determine if a congruent triangle can be constructed using given side measurements.

Component Summary

5.1.1 Warm-Up (~5 min.)	Energy distribution using triangles activity *(SE p.* 223)	5.1.3 Bring It Together (10 min.)	Summarize that the Side-Side-Side (SSS) Congruence Postulate is true *(SE p.* 228)
5.1.2 Exploration (20-25 min.)	Exploration with constructing triangles whose sides are congruent *(SE p.* 224)	5.1.4 Wrap-Up (~5 min.)	Explanation of whether two triangles are congruent given specific side lengths *(SE p.* 229)

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> • Side-Side-Side (SSS) Congruence Postulate <u>Additional Terms:</u> • Angles • Congruence • Rigid motions • Transformation	• Compass • Patty Paper • Ruler • Protractor • (Optional) Highlighters • (Optional) Triangle Strength Video	• This lesson is an in-depth exploration. The design of the lesson is not to give students the information, but for them to discover the SSS Congruence Postulate. Consider doing 5.1.2 Exploration on your own to see how it unfolds to make it easier to connect with students.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may make errors when using the tools for this lesson (compass, patty paper, ruler, and protractor).
Students may overthink creating a second triangle with the same lengths.
Students may not understand how angles are created.
Students may misread side lengths using line segment notation.</td><td>The vast majority of this lesson is 5.1.2 Exploration. Students will be using tools such as a compass and patty paper explicitly during the activity. Consider doing a brief review of the tools prior to launching the lesson if students need that level of support.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
During 5.1.2 Exploration, students will be working together with a partner or small group to determine if two triangles are congruent using side lengths. Some questions like question 3c, ask students to take turns working with different classmates to share their conjectures for creating triangles whose sides are congruent, both explaining and defending their summary.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
In 5.1.2 Exploration, students will be making conjectures based on what they learned through their initial exploration. Based on what they discover, students will create possible arguments that they must share, explain, and defend with other classmates.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For the 5.1.2 Exploration, consider grouping students together by their understanding of the use of geometric tools (particularly) and performance on constructions. This can allow for differentiated support for this lesson based on knowledge needed to complete some of the steps.
5.1.2 Exploration has activities that lead into the discovery that two triangles are congruent if the three sides of one are congruent to the three sides of the other triangle through various entry-points. Consider the use of manipulatives such as popsicle sticks, even when they are not expressly stated, to use in 5.1.2 Exploration if students are not making connections.
5.1.4 Wrap-Up provides an opportunity for students to display their level of understanding of the Side-Side-Side (SSS) Congruence Postulate. The student responses can provide data for which students may need additional support.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.1.1 Warm-Up: STEM

Component Overview: *Students will explore energy distribution using triangles. For this activity, students work independently. There is a video that is available to extend the concept of triangle strength.*

5.1.1 Warm-Up: STEM

Triangles are effective tools for architecture used in the design of buildings and other structures, as they provide strength and stability.

When building materials are used to form a triangle, the design has a heavy base and the pinnacle on the top is capable of handling weight because of how the energy is distributed throughout the triangle. The triangle's use in architecture dates back longer than other common architectural shapes such as the dome, arch, and cylinder. It even predates the wheel.

1. Imagine using straws or toothpicks to make a square or rectangle. What do you think will happen when the engineer puts pressure on the side of the rectangle?
   Answers will vary. When the engineer puts pressure on the side of the rectangle, the rectangle will fall to the side. The rectangle will break.

2. Now imagine you made a triangle using straws or toothpicks. What do you think will happen when the engineer puts pressure on the side of the triangle?
   Answers will vary. When the engineer puts pressure on the side of the triangle, the legs of the triangle will not move. The engineer will have to push harder to break the triangle.

Consider beginning 5.1.1 Warm-Up by reading the narrative language for the whole class to gather buy-in and engagement with the students.
This activity also has a corresponding video that can help engage students in the content of 5.1.1 Warm-Up.

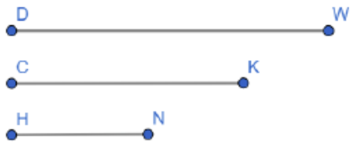
Provide an auditory entry-point by modifying the 5.1.1 Warm-Up activity to a discussion with small groups or the whole class. Questions 1 and 2 allow students to make a guess on what they think will happen. If you do this as a class discussion, it can allow for more ideas to be generated.

5.1.2 Exploration: Are Three Sides Enough?

Component Overview: *Students will explore constructing triangles whose sides are congruent. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss, to ensure that students are on the right track prior to progressing through the next part of the activity.*

5.1.2 Exploration: Are Three Sides Enough?

1. $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ are given. Use the given steps to construct ΔBYT so that its sides are congruent to $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$.



Step 1: Copy $\overline{DW}$. Construct $\overline{BY}$ so that it is congruent to $\overline{DW}$.

Step 2: Copy $\overline{CK}$. Place the compass on point B to create an arc that is the length of $\overline{CK}$.

Step 3: Copy $\overline{HN}$. Place the compass on point Y to create an arc that is the length of $\overline{HN}$.

Step 4: Mark the intersection of the two arcs using a point. Label the point T.

Step 5: Draw $\overline{BT}$ and $\overline{YT}$.

Answers will vary.

Students may not recall the process for creating a line segment. As a refresher, students can refer to Unit 1, Lessons 8 and 9 on copying a line segment.

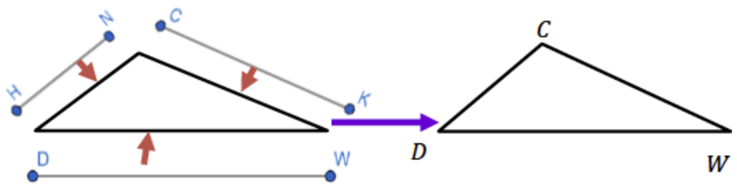
The final image for question 1 is a triangle. Answers may vary with some refection or rotation of the answer shown.

Notice and Wonder

- *What do you notice about your resulting triangle?*
- *What do you wonder about it?*

5.1.2 Exploration: Are Three Sides Enough? (continued)

2. The triangle below was created using the three given line segments.

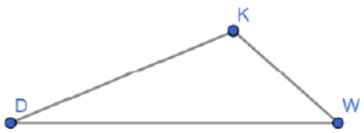


Compare the triangle you constructed with ΔCDW. Describe any similarities or differences.
Answers will vary. ΔCDW was made using the same lines used to make ΔBTY, so it has the same dimensions. Both triangles have sides that are congruent. The interior angle measures are the same. The triangles are located in different places on the plane and are named differently.

3. Maria used a computer program to create line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$. She then moved the line segments to construct ΔDKW. Because the vertices of the line segments $\overline{CK}$ and $\overline{HN}$ coincided with other vertices, she only used the letters D, K, and W.

   a. What do you notice about ΔBYT and ΔDKW?

   Answers will vary. ΔBYT looks the same as ΔDKW. ΔBYT has the same measurements as ΔDKW because it was made with the same lines.



Discussion prompts for question 2:
- *How can you determine that both triangles have the same dimensions?*
- *How do you know if the triangles have sides that are congruent to one another?*

What are the benefits of using a computer program to create line segments? How could you confirm the accuracy of the line segments made from that method?

For question 3a, allow different students to share out what they noticed to check for similarities and potential misconceptions that need to be addressed.

5.1.2 Exploration: Are Three Sides Enough? (continued)

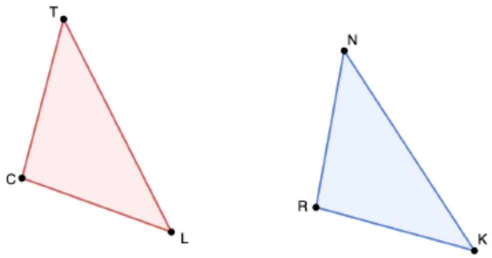
b. The line segments $\overline{DW}$, $\overline{CK}$, and $\overline{HN}$ were used to construct three different triangles. Work with your partner to write a conjecture about creating triangles whose sides are congruent.

Answers will vary. If a triangle is constructed using the same side lengths, the new triangle will be congruent to the original.

c. Ask two classmates about their conjectures. Record their conjectures and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Conjecture for Creating Triangles Whose Sides are Congruent	My Summary of Their Reason	Initials
Answers will vary.		
Answers will vary.		

4. Use patty paper to determine if ΔCTL and ΔRNK were created using the same three line segments.



Answers will vary. Yes, ΔCTL and ΔRNK line up perfectly.

For question 3b, listen carefully for students making conjectures that aren't rooted in evidence. For example, some students may make conjectures that are based on their opinion but not based on what they have observed in 5.1.2 Exploration.

For question 3c, consider reviewing the instructions with the students to have a better idea of what is expected from this activity.

For a challenge, have students find one additional student who has a conjecture they agree with and one student who has a conjecture they don't agree with. For both cases, have students explain why they do or don't agree with that conjecture.

Students may need additional support when it comes to the use of patty paper. Consider providing a brief tutorial for students on the use of patty paper prior to releasing them to work on question 4.

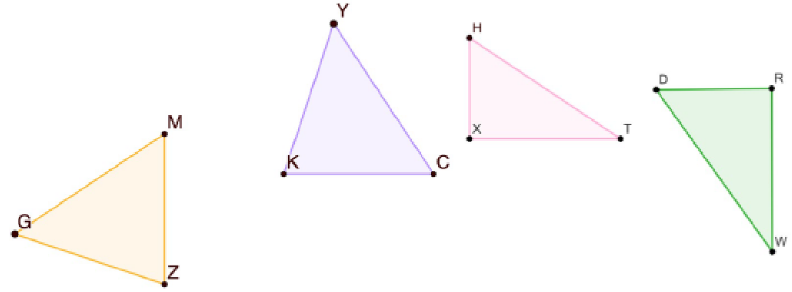
5.1.2 Exploration: Are Three Sides Enough? (continued)

5. Discuss the following with your partner: If three sides of a triangle are congruent, does that mean all of the angles of the triangles are congruent too? Explain your decision and state the evidence used for the decision.

   Answers will vary. Yes, all angles are congruent. If three sides of a triangle are congruent, then all the angles of the triangles are congruent because each pair of sides will create the same angles in order to fit together to make a triangle.

6. Two sets of triangles are shown.

   ΔMGZ and ΔKYC $\quad$ ΔHTX and ΔWDR



   a. Use patty paper to determine if the sides of the two triangles are congruent.
   b. $\underline{\Delta\ MGZ}$ and $\underline{\Delta CYK}$ are congruent because all of the sides are congruent.
   c. For the two triangles that are congruent, use patty paper to verify that the angles are congruent.
   d. For the two triangles that are congruent, list the sides on each triangle that are congruent to each other.

   $\overline{ZG} \cong \overline{KY}$, $\overline{CK} \cong \overline{MZ}$, and $\overline{CY} \cong \overline{GM}$

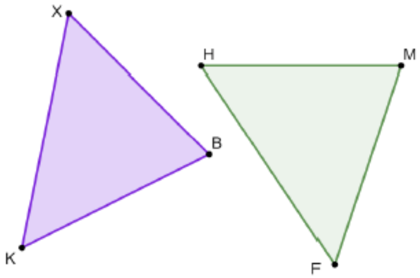
Notice and Wonder

- *What do you notice about the two sets of triangles presented in question 6?*
- *What are some things that you wonder about these triangles?*

When reviewing with the class, consider allowing a student to come to the board or document camera to show how they used the patty paper to determine that corresponding sides and angles of the two triangles were congruent.

5.1.2 Exploration: Are Three Sides Enough? (continued)

7. ΔKXB and ΔHFM are shown. $HM = 4.2$, $XB = 4.2$, $XK = 5$, and $KB = 4.4$.



a. Discuss with your partner if the given measurements are sufficient to determine whether ΔKXB and ΔHFM are congruent. Explain your decision.

Answers will vary. The given measurements are not sufficient to determine if the triangles are congruent because we do not know if all sides are the same length. We only know that $\overline{XB} \cong \overline{HM}$.

b. Use patty paper, a protractor, a ruler, or a compass to measure all of the parts of ΔKXB and ΔHFM. Then, complete the sentence.

ΔKXB and ΔFHM are [○ congruent / ○ not congruent] because [○ 0 / ○ 1 / ○ 2 / ○ 3] side(s) and [○ 0 / ○ 1 / ○ 2 / ○ 3] angle(s) are congruent.

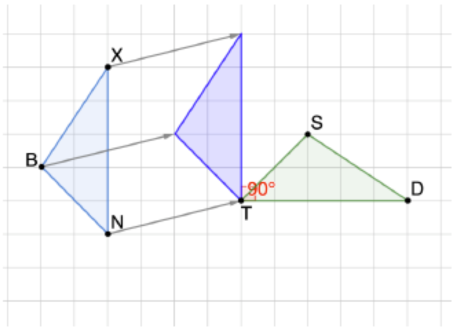
Answer: congruent; 3; 3

Consider using colored highlighters or markers to illustrate the connection between the side lengths that are congruent to one another.

What process did you and your partner go through to decide whether the given measurements are sufficient for determining if the triangles are congruent?

- What tool(s) did you use to complete the statement in question 7b?
- Why did you choose that tool?

8. A series of transformations shows how $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated, then rotated $90°$ about point T to create $\overline{SD}$, $\overline{DT}$, and $\overline{TS}$.



a. Use the grid to determine if $BN = TS$, $BX = DS$, and $XN = TD$.
Answers will vary. Yes, all corresponding sides of the triangle are congruent.

b. Discuss with your partner how the transformations that were used ensure that the length of each line segment was preserved.

Answers will vary. The transformations that were used only changed the location and not the length of each side of the triangle. Therefore, the transformations were rigid transformations and preserved the lengths of all sides of the triangle.

c. Use the word bank to complete the statements. Words may be used more than once.

Word Bank				
angles	congruent	congruence	parts	rigid motions

Since $\overline{BX}$, $\overline{XN}$, and $\overline{BN}$ were translated then rotated $90°$, the length of each line segment was preserved because of the definition of congruence in terms of ___*rigid motions*___. The definition of congruence in terms of _____*rigid motions*____ also ensures that the _*angles*_ are congruent. Because all of the __*parts*___ of ΔXNB are ___*congruent*___ to all of the corresponding _*parts*____ of ΔDTS, the two triangles are ___*congruent*___.

Prompts to consider for the image of question 8:
- *What do you notice about the images?*
- *What do you wonder about the images?*
- *What do you know about translations?*
- *What do you know about rotations?*

In question 8b, students may not recall the term "preserved" from Unit 1 Lesson 2 and may need additional support. Consider defining what this word means within the context prior to releasing students to work on the question.
For question 8b, students are not required to write down their answers. They will use the discussion from question 8b to complete the statement in question 8c.
For question 8c, let students know that word choices can be used more than once throughout the statement.
Consider reviewing question 8c during the 5.1.3 Bring It Together.

5.1.3 Bring It Together: Triangle Congruence

Component Overview: *Students will explain, using what they learned in 5.1.2 Exploration, that the Side-Side-Side (SSS) Congruence Postulate is true. For this activity, students should work with their partner to complete the questions.*

5.1.3 Bring It Together: Triangle Congruence

Side-Side-Side (SSS) Congruence Postulate

Two triangles are congruent if the three sides of one are congruent to the three sides of the other, respectively.

1. Work with your partner to explain how the exploration has shown that the Side-Side-Side (SSS) Congruence Postulate is true.

   Answers will vary. The explorations showed us Side-Side-Side is true by using three different segments to create congruent triangles. The triangles with three congruent sides also had congruent angles, so three congruent sides were enough to show the triangles were congruent.

Students may need a refresher on the term "postulate" and how it applies to Side-Side-Side.

After giving students time to discuss, consider asking students to share with the class what they discussed with their partner. Also, allow other students to critique their reasoning and provide feedback.

5.1.4 Wrap-Up: Cool Down

Component Overview: *Students will explain whether two triangles are congruent given specific side lengths. Students should complete this activity independently so that accurate student-level data on their understanding of the lesson content can be collected.*

5.1.4 Wrap-Up: Cool Down

1. Maggie and Naomi both constructed a triangle using the side lengths shown.

   - $CN = 2.5$ inches (in.)
   - $NK = 3.75$ in.
   - $CK = 5.25$ in.

   Explain whether it is possible to find out if the two triangles Maggie and Naomi created will be congruent.

   Answers will vary. Since Maggie and Naomi created triangles from the same three side lengths, their triangles will be congruent by SSS Congruence Postulate.

This question assesses student understanding of the Side-Side-Side (SSS) Congruence Postulate. The data from the student response can provide insight on whether or not the students need additional support on the content from this lesson.
When working with students in small groups who struggled with the Ticket Out the Door, provide students with paper strips with three lengths and ask them to create a triangle. Then, allow each member of the small group to share what they notice and wonder about the triangles.

5.2 Exploring Side-Angle-Side and Angle-Side-Angle Triangle Congruence

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.5.1 Construct a copy of a segment or an angle.	Learning Targets	• I can use constructions to determine if SAS congruence can be used to show that two triangles are congruent. • I can use constructions to determine if ASA congruence can be used to show that two triangles are congruent. • I can use constructions to determine if SSA congruence can be used to show that two triangles are congruent.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I use constructions to determine if SAS, ASA, and SSA congruence can be used to show that two triangles are congruent? • How can I use given information to determine if SSS, SAS, or ASA can be used to show that two triangles are congruent?
Lesson Narrative	In this lesson, students will explore congruent relationships using side lengths and specified angles to determine if two triangles are congruent. Students will use their explorations with constructions to discover the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem. They will use this knowledge to determine if two triangles are congruent.

Component Summary	5.2.1 Warm-Up (~5 min.)	Geometry in art *(SE p.* 230)	5.2.3 Bring It Together (10-15 min.)	Formalized introduction to the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem *(SE p.* 232)
	5.2.2 Exploration (20-25 min.)	Exploring various constructions to discover congruent relationships between triangles *(SE p.* 230)	5.2.4 Wrap-Up (~5 min.)	Practice using the triangle congruence criterion to determine if triangles are congruent *(SE p.* 233)

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Side-Angle-Side (SAS) Congruence Postulate • Angle-Side-Angle (ASA) Congruence Theorem Additional Terms: • Angles • Congruence	• Compass • Ruler • Patty Paper	• The design of the lesson is not to give students the information, but for them to discover the SAS Congruence Postulate and ASA Congruence Theorem. Consider doing the exploration on your own to see how this unfolds to make it easier to facilitate with students.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may think that SSA can be used to determine congruency between two triangles. • Students may forget the “included” component of SAS and ASA. • Students may misread side lengths using line segment notation.	• The majority of this lesson is 5.2.2 Exploration. Students will be using tools, such as a compass and patty paper explicitly during the activity.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share For 5.2.2 Exploration, students will be working together to discover congruent relationships between sides and angles. Students will get the opportunity to work on their own, pair up with a partner, and share their work with one another to come to a consensus.	**MA.K12.MTR.4.1: Discussion and Reflection** In 5.2.3 Bring It Together, students will be making conjectures based on what they learned through their initial exploration. Students will create arguments based on what they explored in the prior activity.
Differentiate Instruction	• Consider pairing students based on their performance in the prior lesson. The data from 5.1.4 Wrap-Up from Lesson 1 can serve as data for student groupings for this lesson. • In 5.2.3 Bring It Together question 2, have students mark the congruent parts on the given triangles. Then, have the student write the words “side” or “angle” next to the marked part. This move will help students visualize which congruence postulate/theorem is represented in the problem. • 5.2.4 Wrap-Up provides students with an opportunity to demonstrate mastery of determining if two triangles are congruent, including what theorem or postulate was used to make that determination. Consider having students list what side/angle relationships are valid for determining congruence of triangles and what relationships aren’t for an additional check for understanding.	
Lesson Notes:		

5.2.1 Warm-Up: Geometry in Art

Component Overview: *Students will explore geometry in art through the largest Pysanka. This activity is designed to be done independently.*

5.2.1 Warm-Up: Geometry in Art

A Pysanka is a decorated egg that is traditional in Ukraine. The decoration on the eggs is created using a wax resist (batik) method. Families often have their own unique pattern. The largest Pysanka is in Vegreville, Canada. It weighs 5,000 pounds, is 31 feet (ft) tall, and 18 ft wide. A traditional Pysanka egg is 2.5 inches tall and 1.75 inches wide.

1. What do you notice about Pysanka eggs?
   Answers will vary. The Pysanka egg is large. I see that the egg has been decorated with many shapes.

2. Estimate the number of traditional Pysanka eggs that will fit into the largest Pysanka. Show your work or explain how you found your estimate.
   Answers will vary. Estimated values: Traditional egg 3 inches tall and 2 inches wide. This large egg is 375 inches tall and 220 inches wide. Using the tall dimension 375 / 3 = 125, using the wide dimension 220 / 2 = 110. To estimate, I should use the smaller number, so I estimate that about 110 traditional Pynsanka eggs will fit inside the largest Pysanka egg.

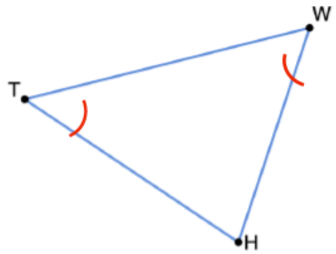
Allow students to share their answers to question 2 with the entire class. Encourage students to agree or disagree with student estimates as viable or not viable.

5.2.2 Exploration: Congruence Relationships

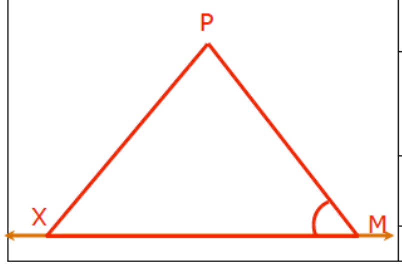
Component Overview: *Students will complete various constructions to discover congruent relationships between triangles. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss, in order to ensure students are on the right track prior to progressing through the next part of the activity.*

5.2.2 Exploration: Congruence Relationships

1. Triangle TWH is shown.



   a. Construct ΔXMP so that $\overline{XM} \cong \overline{TW}$, $\overline{PM} \cong \overline{HW}$, and $\angle TWH \cong \angle PMX$ using the given steps.

Construction	Steps
	1. Copy $\overline{TW}$ using a compass to construct $\overline{XM}$ using the orange line so that $\overline{XM} \cong \overline{TW}$.
	2. Copy $\angle TWH$ to construct $\angle PMX$ so that $\angle TWH \cong \angle PMX$.
	3. Copy $\overline{HW}$ to construct $\overline{PM}$ so that $\overline{PM} \cong \overline{HW}$.
	4. Draw $\overline{PX}$.

   b. Use patty paper to determine if ΔXMP is congruent to ΔTWH.

   c. Discuss the following with your partner: If two sides of a triangle and their included angles are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.
   Answers will vary. If two sides of a triangle and their included angle are congruent, then all parts of the triangles are congruent. After using the patty paper, I can see that the remaining side is congruent to its corresponding side and the other two angles are congruent to their corresponding angles. This makes all sides and angles of the triangle congruent to the corresponding parts.

For this activity, students will need the following materials:

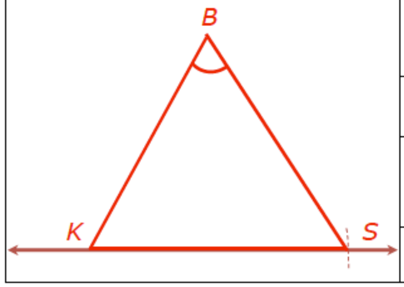
- *Compass*
- *Ruler*
- *Protractor*
- *Patty Paper*

For each question, consider using technology, such as a document camera or GeoGebra to illustrate the steps of the construction so that students can follow along in review.

- *When constructing $\angle PMX$, how long do you draw $\overrightarrow{MP}$?*
- *How can you use patty paper to determine if all parts of ΔXMP are congruent to ΔTWH?*
- *What does "included angle" mean?*

5.2.2 Exploration: Congruence Relationships (continued)

2. Construct ΔKSB so that $\overline{KS} \cong \overline{TH}$, $\overline{SB} \cong \overline{TW}$, and $\angle TWH \cong \angle SBK$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{KS}$ using the red line so that $\overline{TH} \cong \overline{KS}$.
	2. Copy $\overline{TW}$ to construct $\overline{SB}$ so that $\overline{TW} \cong \overline{SB}$.
	3. Copy $\angle TWH$ to construct $\angle SBK$ so that $\angle TWH \cong \angle SBK$.
	4. Draw $\overline{BK}$.

a. Use patty paper to determine if ΔKSB is congruent to ΔHTW.

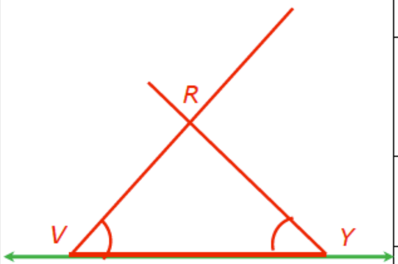
b. Discuss the following with your partner: If two sides of a triangle and a non-included angle are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.
Answers will vary. No, it does not. If two sides of a triangle and a non-included angle are congruent, then it cannot be determined if all parts of the triangle are congruent. The non-included angle doesn't provide any information about the 3rd side length and the triangle can be created different ways using two sides and a non-included angle.

The ΔTWH is not on every question in the student workbook. Consider having students use patty paper to cut down on having to flip to previous pages when working through the additional questions.
Students may struggle with knowing what angle to draw $\overline{SB}$ at point S. Resist giving them help here, as this is a key part of determining that SSA is not a valid statement of congruence.

Why do you think that if two sides of a triangle and a non-included angle are congruent then we can't know for certain that all parts of the triangles are congruent too?

Why do you think the included vs. non-included angles would make a difference in the determination of whether not two triangles are congruent?

3. Construct ΔRYV so that $\overline{VY} \cong \overline{TH}$, $\angle WTH \cong \angle RVY$, and $\angle WHT \cong \angle RYV$ using the given steps.

Construction	Steps
	1. Copy $\overline{TH}$ using a compass to construct $\overline{VY}$ using the green line so that $\overline{TH} \cong \overline{VY}$.
	2. Copy $\angle WHT$ to construct $\angle RYV$ so that $\angle WHT \cong \angle RYV$.
	3. Copy $\angle WTH$ to construct $\angle RVY$ so that $\angle WTH \cong \angle RVY$.
	4. Label the intersection of the sides of the two angles point R.

a. Use patty paper to determine if ΔRYV is congruent to ΔWHT.

b. Discuss the following with your partner: If two angles of a triangle and their included side are congruent, does that mean all of the parts of the triangles are congruent too? Explain your decision and state the evidence used for the decision.
Answers will vary. If the congruent side is between the congruent angles, all parts of the triangle will be congruent. With this triangle construction, there will be no other way to create a triangle so it must be congruent.

When reviewing the answers, consider having a student come up to the board to demonstrate the steps of the construction in question 3.

How long should the rays of the copied angles be drawn?

- *What are the similarities and differences between the three triangles you constructed in this activity?*
- *Is there any relationship between the information provided for each triangle?*
- *What do you notice about these three?*
- *What do you wonder about them?*

5.2.3 Bring It Together: Triangle Congruence

Component Overview: *Students will connect what they discovered in 5.2.2 Exploration to a formalized introduction to the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem.*

5.2.3 Bring It Together: Triangle Congruence

1. Complete the statements.

   a. ΔXMP was created using [○ 1 / ○ 2 / ○ 3] side(s) and [○ 1 / ○ 2 / ○ 3] angle(s).

   The construction showed that it is [○ possible / ○ not possible]

   to assume that if the [○ 1 / ○ 2 / ○ 3] side(s) of a triangle

   and their [○ adjacent angle(s) / ○ included angle(s) / ○ opposite angle(s)] are congruent,

   then the triangles are congruent.

   b. ΔKSB was created using [○ 1 / ○ 2 / ○ 3] side(s) and [○ 1 / ○ 2 / ○ 3] angle(s). The construction showed that it is

   [○ possible / ○ not possible] to assume that if the [○ 1 / ○ 2 / ○ 3]

   side(s) of a triangle and the [○ included angle(s) / ○ non-included angle(s)]

   are congruent, then the triangles are congruent.

Consider modeling question 1a as a think-aloud with the class. Students often need support on how to think through fill-in-the-blank style questions.

For question 1b, consider having students review their answers as a class prior to moving forward to question 1c. This way, students can engage in question 1c as an independent practice for themselves.

What are the differences between what you discovered in question 1a and 1b?

5.2.3 Bring It Together: Triangle Congruence (continued)

c. ΔRYV was created using
- 1
- 2
- 3

side(s) and
- 1
- 2
- 3

angle(s).

The construction showed that it is
- possible
- not possible

to assume that if the
- 1
- 2
- 3

side(s) of a triangle and its
- adjacent angle(s)
- included angle(s)
- opposite angle(s)

are congruent, then the triangles are congruent.

Side-Angle-Side (SAS) Congruence Postulate

If two sides and the included angle of one triangle are congruent to two sides and the included angle of a second triangle, respectively, then the two triangles are congruent.

Angle-Side-Angle (ASA) Congruence Theorem

If two angles and an included side of one triangle are congruent respectively to two angles and an included side of a second triangle, then the two triangles are congruent.

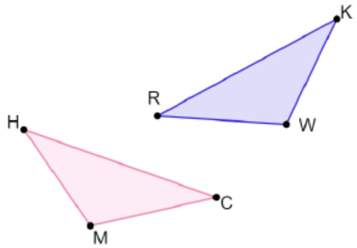
What makes the triangle in question 1c similar to the ones in questions 1a and 1b? What are some of the differences?

When reviewing the Side-Angle-Side (SAS) Congruence Postulate and the Angle-Side-Angle (ASA) Congruence Theorem, consider reviewing them within the context of the work students have already done. For example, don't simply give the postulate, but connect it to the corresponding triangle from the prior three they have worked with as well.

Why do you think that there is an SAS Congruence Postulate and an ASA Congruence Theorem but no congruence postulate or theorem for SSA?

5.2.3 Bring It Together: Triangle Congruence (continued)

2. Triangles HCM and RKW are shown. Determine if the given information is sufficient to determine whether the triangles are congruent. If so, state the congruence theorem or postulate to show that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{MC} \cong \overline{RW}$, $\overline{HC} \cong \overline{KR}$, $\angle MCH \cong \angle WRK$	Yes	SAS
$\overline{MC} \cong \overline{KW}$, $\overline{HM} \cong \overline{WR}$, $\overline{HC} \cong \overline{KR}$	Yes	SSS
$\overline{KW} \cong \overline{HM}$, $\angle KRW \cong \angle CHM$, $\overline{MC} \cong \overline{RW}$	No	
$\overline{RK} \cong \overline{CH}$, $\angle MCH \cong \angle WRK$, $\angle MHC \cong \angle RKW$	Yes	ASA
$\angle RKW \cong \angle CHM$, $\angle RWK \cong \angle CMH$, $\overline{KW} \cong \overline{HM}$	Yes	ASA

For each line of the chart, students can mark the congruent parts on the given triangles. Then, have the student write the words "side" or "angle" next to the marked part. This move will help students visualize which congruence postulate/theorem is represented in the problem. Teachers can also use this step to check for where misunderstandings occur when developing remediation plans.

To add some physical movement to this activity, consider labeling each corner of the room with the following:

- *SAS*
- *ASA*
- *SSS*
- *SSA*

Display each row of the table and have students choose a corner to determine which one they agree with. Have students share out their rationale for their choice.

After completing the chart, ask students why SSA was never an answer for the third column.

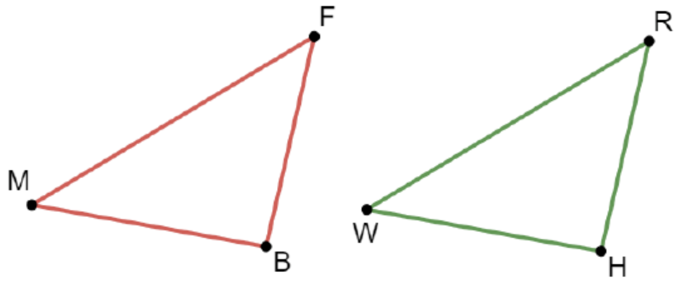
For a challenge, have students create two triangles that could be proved to be congruent by a specific theorem or postulate.

5.2.4 Wrap-Up: Ticket Out the Door

Component Overview: *Students will use the triangle congruence criterion they have learned to determine if triangles are congruent. This activity should be done independently to determine mastery of the content of this lesson.*

5.2.4 Wrap-Up: Ticket Out the Door

Triangles BFM and WRH are shown.



1. Determine if the given information is sufficient to determine that the triangles are congruent. If so, state the congruence theorem or postulate to show that.

Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{FB} \cong \overline{HW}$, $\overline{FM} \cong \overline{RW}$, $\angle BFM \cong \angle HWR$	*Yes*	*SAS*
$\overline{FB} \cong \overline{RH}$, $\angle FBM \cong \angle RHW$, $\overline{MB} \cong \overline{WH}$	*Yes*	*SAS*
$\overline{FB} \cong \overline{HW}$, $\overline{BM} \cong \overline{HR}$, $\overline{FM} \cong \overline{RW}$	*Yes*	*SSS*

This question assesses student understanding of the Side-Side-Side (SSS) Congruence Postulate and Side-Angle-Side (SAS) Congruence Postulate. The data from the student responses can provide insight on whether the students need additional support on the content from this lesson.

Consider having students list what side/angle relationships are valid for determining congruence of triangles and what relationships are not, for an additional check for understanding.

5.3 Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.5.1 Construct a copy of a segment or angle.	Learning Targets	• I can use constructions to determine if AAS congruence can be used to show that two triangles are congruent. • I can determine if HL congruence can be used to show that two triangles are congruent using deduction.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I use constructions to determine if AAS congruence can be used to show that two triangles are congruent? • How can I determine if HL congruence can be used to show that two triangles are congruent?
Lesson Narrative	In this lesson, students will continue their use of constructions to explore additional conditions for determining the congruence of two triangles. Students will use constructions to explore Angle-Angle-Side and Hypotenuse-Leg triangle congruence.

Component Summary	5.3.1 Warm-Up (~5 min.)	Reflection *(SE p.* 234)	5.3.4 Bring It Together (10 min.)	Formalized introduction to the Angle-Angle-Side (AAS) Congruence Theorem and the Hypotenuse-Leg (HL) Theorem *(SE p.* 239)
	5.3.2 Exploration (10-15 min.)	Exploration with different constructions to determine triangle congruency *(SE p.* 235)	5.3.5 Try It (5-10 min.)	Determining if two triangles are congruent based on given information and which postulate or theorem would be used *(SE p.* 240)
	5.3.3 Collaboration (10 min.)	Collaboration on reasoning arguments for the criteria for triangle congruence *(SE p.* 237)	5.3.6 Wrap-Up (~5 min.)	Self-reflection on the understanding of each congruence postulate or theorem *(SE p.* 240)

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Angle-Angle-Side (AAS) Congruence Theorem • Hypotenuse-Leg (HL) Theorem Additional Terms: N/A	• Patty paper	• The design of the lesson is not to give students the information, but for them to discover the AAS Congruence Theorem and the Hypotenuse Leg Theorem. Consider doing the Exploration on your own to see how this unfolds to make it easier to connect with students.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not use the included angle when appropriate.
Students may forget to use the Hypotenuse Leg theorem as an option for proving congruence.
Students may misread side lengths using line segment notation.</td><td>This lesson is primarily an exploration. Students will be using tools such as patty paper explicitly during the activity. Consider doing a brief review of what students did in the prior lessons as a refresher prior to beginning this lesson.
If time is an issue, consider assigning 5.3.5 Try It for homework.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Clarify, Critique, Correct
During 5.3.3 Collaboration, students analyze, reflect on, and improve the sample arguments on triangle congruence. When writing the improvements, students will clarify and add additional information to make the correct argument more convincing.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
In 5.3.4 Bring It Together, students will be working to bring together all they learned in 5.3.2 Exploration to make a conjecture and learn the formalized definitions of Angle-Angle-Side and Hypotenuse-Leg.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider pairing students based on their performance in the prior lesson. The data from the Wrap-Up activity from Lesson 1 can serve as strong data for student groupings for this lesson.
Consider the extended use of the patty paper when showing congruence throughout the lesson. Consider extended exploration time with students in small groups who need additional support.
5.3.5 Try It provides students with an opportunity to demonstrate mastery of determining if two triangles are congruent, including what theorem or postulate was used to make that determination. Consider using the student’s labeling of the given information and the naming of the parts on the triangles as data when determining the level of mastery of congruent triangles.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.3.1 Warm-Up: Giving Back

Component Overview: *Students explore a case study on working through challenges to succeed and giving back.*

5.3.1 Warm-Up: Giving Back

Morgan State University is a historically black, public research university in Baltimore, Maryland. In the early 1960s, Calvin E. Tyler Jr., a young African American man who was the first in his family to go to college, attended Morgan State. However, in 1963 he was forced to drop out when he was unable to pay his tuition.

He began working as a truck driver for the United Parcel Service (UPS). Throughout his career at UPS, he continued to be promoted and ultimately served as both the senior vice president for U.S. operations and as a director within the company. On February 23, 2021, he and his wife Tina pledged $20 million to Morgan State University's endowment to go toward scholarships for students in financial need.

Choose one of the following questions to answer.

1. How do you think having a strong drive to succeed likely played a role in Mr. Tyler's career?
   Answers will vary.

2. Why do you think it was important to Mr. Tyler to pledge this endowment?
   Answers will vary.

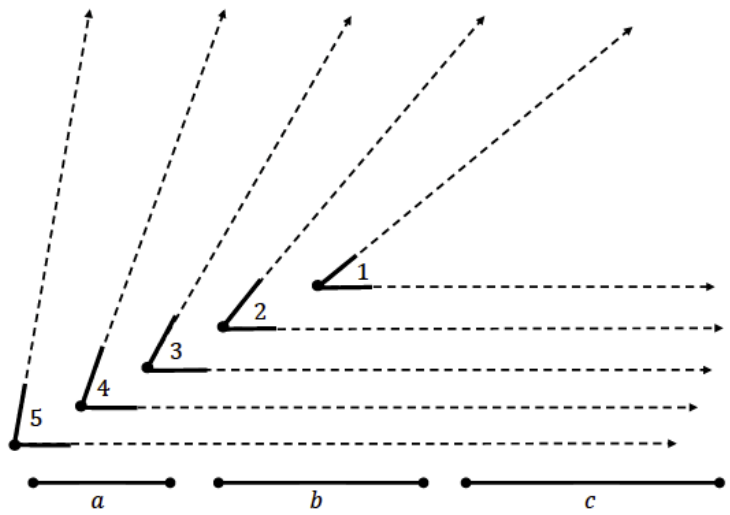
- *What obstacles have made math challenging for you so far?*
- *What connections are you making between Mr. Tyler's career and your journey in mathematics?*

5.3.2 Exploration: Which Other Criteria Is Enough?

Component Overview: *Students will explore different constructions to determine using new criteria for triangle congruency. For this activity, consider having each partner pair or small group work on one question at a time before bringing the class back together to discuss what they did, in order to ensure students are on the right track prior to progressing through the next part of the activity.*

5.3.2 Exploration: Which Other Criteria is Enough?

Five angles labeled 1-5 and three line segments labeled $a, b,$ and c are given.



Use the given parts to create triangles by copying the parts needed onto patty paper using the given criteria below.

Helpful Hint: When copying an angle, include some (or all) of the dotted lines so that angles and line segments can be connected.

For this activity, consider providing students with a review of the instructions and setup for this problem. Do a verbal review of the angles and line segments provided on this page.

Notice and Wonder

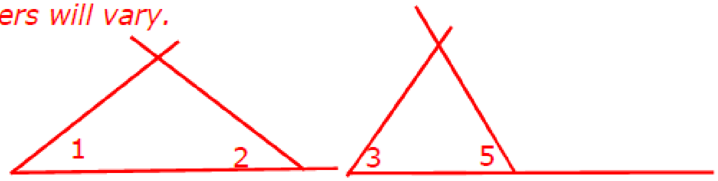
- *What do you notice about each of the angles provided? What about the line segments?*
- *What do you wonder about each of the angles provided? What about the line segments?*

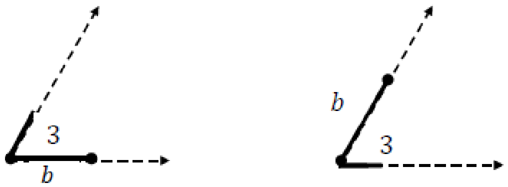
5.3.2 Exploration: Which Other Criteria Is Enough? (continued)

1. Construct two different triangles using each set of criteria. Use the provided steps in each criteria.

Criteria 1: Given two angles (AA)
Step 1: Copy $\angle 1$ onto a piece of patty paper.
Step 2: Close the triangle using $\angle 2$.
Step 3: Trace the completed triangle. Label $\angle 1$ and $\angle 2$ in the completed triangle.
Step 4: Label your triangle "Criteria 1 (AA)."
Step 5: Repeat these steps with $\angle 3$ and $\angle 5$.

Answers will vary.



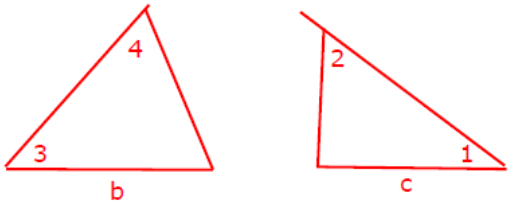
Criteria 2: Given two angles and a non-included side (AAS)
Step 1: Copy side b onto a piece of patty paper. Label side b on your patty paper.
Step 2: Place $\angle 3$ at one of the ends of side b and trace the ray that does not overlap with side b to copy the angle. Label $\angle 3$. 
Step 3: Use $\angle 4$ to close the triangle such that $\angle 4$ is opposite from side b. Label $\angle 4$.
Step 4: Trace the completed triangle. Label your triangle "Criteria 2 (AAS)."
Step 5: Repeat these steps with side c, $\angle 1$, and $\angle 2$.

Avoid assisting students with locating the placement of $\angle 2$ on the side of $\angle 1$ (likewise for $\angle 5$ and $\angle 3$). Students should naturally have different placements of this second angle since the length of the side of the first angle is not specified.

Teachers may want students to tape their patty paper below each criterion.

What are some things that you noticed about your triangles in Criteria 1?

Consider walking students through Criteria 2. The instructions may confuse students in the process. Guide students through each step, one-by-one, modeling each step and ensuring that students are completing each step correctly.



2. Compare your four triangles with your partner's.
3. Write your partner's name on the line in the table. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	∠1 and ∠2	∠3 and ∠5	side *b*, ∠3, and ∠4	side *c*, ∠1, and ∠2
My triangle is congruent to __Partner_'s triangle.	Answers will vary. False	Answers will vary. False	True	True

4. With your partner, find another pair of students to form a group of four. Write the two new students' names in the table below. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	∠1 and ∠2	∠3 and ∠5	side *b*, ∠3, and ∠4	side *c*, ∠1, and ∠2
My triangle is congruent to __Student_'s triangle.	Answers will vary. False	Answers will vary. False	True	True
My triangle is congruent to __Student_'s triangle.	Answers will vary. False	Answers will vary. False	True	True

Students are not writing for question 2; they're just discussing the comparison. Their discussion will be summarized in the tables for questions 3, 4, and 5. The answer written here is guidance for what to listen for while monitoring the student discussion. The triangles in Criteria 1 are the same shape but different sizes. Highlighting this as a "notice" will build on the future learning of triangle similarity with AA.

Discussion questions for question 2:
- *What is similar about all the triangles that follow Criteria 1?*
- *What did you notice about the triangles created under Criteria 2?*
- *What is the relationship between the triangles in Criteria 1 and Criteria 2?*

Consider using physical movement in this process by having students move about the classroom in order to find an additional student pair.

5.3.2 Exploration: Which Other Criteria Is Enough? (continued)

5. Discuss with your group: If given either set of criteria, does that mean all of the parts of the triangles are congruent? Then, write your explanation in the table.

Criteria	Explanation
Given two angles (AA)	*Answers will vary. The triangles created with two congruent angles are not necessarily congruent. The triangles can be created different ways with different side lengths.*
Given two angles and non-included side (AAS)	*Answers will vary. If two adjacent angles in a triangle are congruent, then their third angle is also congruent because all triangles have a sum of $180°$. If the triangles, then also have a corresponding side that is congruent, they will be congruent by ASA Congruence Theorem.*

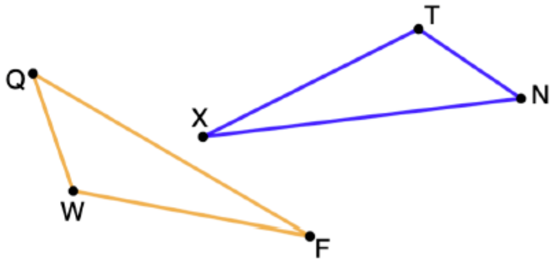
Allow students to share with the entire class their explanation. Have students decide if they agree or disagree with the explanation, clarifying and refining their classmates' explanations when necessary.

5.3.3 Collaboration: Making Connections

Component Overview: *Students will work together to make connections with criteria for triangle congruency. For this activity, release students to work on the activity with a partner, without extensive prompting.*

5.3.3 Collaboration: Making Connections

1. ΔQWF and ΔNTX are shown. $\overline{FQ} \cong \overline{XN}$, $\angle FQW \cong \angle XNT$, and $\angle QWF \cong \angle NTX$.



Two students discussed the information given.

Mabel thinks there is not enough information to determine congruence, because the side given is not between the two angles given.	Xavier thinks the triangles are congruent because he knows that the third angles in the triangles have to be congruent, and then there would be a side between two angles.

With your partner, discuss both students' reasoning. Then, answer the following.

a. Explain whether you agree with Mabel or Xavier.
Answers will vary. Xavier is correct. Xavier understands that the side does not have to be included between the two angles to determine congruence.

b. What additional information could be added to the argument you agree with to make it more convincing?
Answers will vary. Xavier could continue to explain that if two angles are congruent, the third must also be congruent. Knowing this he may point out that any triangle with 2 congruent angles and a congruent side will either fall into AAS or ASA. Both are able to prove congruence.

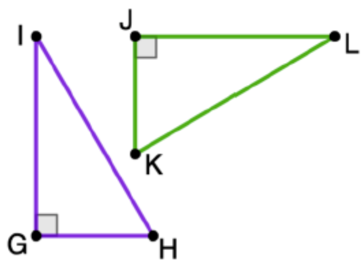
Consider allowing students to share where they believe the thinking for both Mabel and Xavier is coming from. Have students vote on whether they agree with Mabel or with Xavier.

- *Where does Mabel have an error in her thinking?*
- *If she asked you for help, what information would you tell her to help correct her?*

- *How do you know additional information is needed for Xavier's explanation?*
- *How would you know what specific additional information would be needed?*
- *How would that strengthen his case?*

5.3.3 Collaboration: Making Connections (continued)

2. Triangle IHG and ΔLKJ are shown. $\overline{IH} \cong \overline{LK}$, $\overline{HG} \cong \overline{KJ}$, and $\angle IGH$ and $\angle LJK$ are right angles. Eli and Kaylah had the following discussion. Eli said, "In a situation where you have two sides without the angle in between them, you can't tell if the triangles are congruent." Kaylah said, "Normally I would agree with you, except these are right triangles."



   a. Discuss with your partner what Kaylah might be thinking. Write a summary of your discussion.
   Answers will vary. Kaylah realizes that because the triangles are right triangles the Pythagorean Theorem can be used to find the unknown third sides and thus show they are congruent.

   b. When given two sides of a right triangle, how can the length of the third side be determined?
   By using the Pythagorean Theorem $a^2 + b^2 = c^2$

- *How does Kaylah's opinion differ from that of Eli?*
- *Why would she think that the triangles being right triangles would cause a difference in Eli's definition?*

Consider leveraging an Agree or Disagree instructional protocol. First, have students determine whether they agree or disagree with Kaylah's argument. Have students share out their rationale for why they chose to agree or disagree with Kaylah.

For an additional challenge, have students write a paragraph proof showing the two triangles are congruent from the given information.

Kaylah continued, "For a right triangle, as long as I have two of the sides, I can find the length of the third side. Then I can use SSS since all three corresponding sides would be congruent."

c. Work with your partner to explain why Kaylah's argument could work to show congruence for any pair of right triangles where the hypotenuse and one leg are identified as congruent.
Answers will vary. Kaylah would be correct because the Pythagorean Theorem can be used to find the missing side. Since the other two sides of the triangle are congruent, the third side will also be congruent allowing us to connect with SSS congruence postulate.

d. Explain whether Kaylah's argument would be needed to show congruence between right triangles where both legs of the triangles are identified as congruent.
Answers will vary. No Kaylah's argument would not be needed to show congruence between the right triangles. Given congruence of both legs of the triangle and the included right angle we can prove congruence under SAS.

If needed, consider probing student thinking as they collaborate to make connections:

- *How does Kaylah's reasoning about right triangles relate to SSS Congruence?*
- *How does Kaylah's reasoning about right triangles relate to SAS Congruence?*

Listen as students discuss questions 2c and 2d to determine misconceptions that may need to be addressed as well as student ideas to be shared with the whole group. Consider facilitating a whole group discussion to wrap up these ideas and address misconceptions prior to moving on to the next component.

5.3.4 Bring It Together: Triangle Congruence

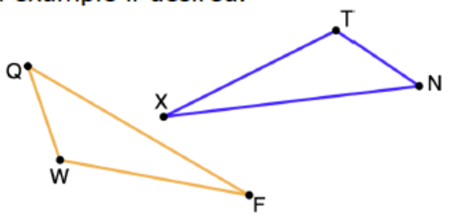
Component Overview: *Students will connect what they have learned through their exploration to the formal definition of the Angle-Angle-Side (AAS) Congruence Theorem and the Hypotenuse-Leg (HL) Theorem. For this activity, provide the narrative of the AAS definition and release students to work with their partners on questions 1 and 2. Provide the narrative of the Hypotenuse-Leg Theorem and release students to complete question 3.*

5.3.4 Bring It Together: Triangle Congruence

Angle-Angle-Side (AAS) Congruence Theorem

If two angles and a side opposite one of them are congruent to two angles and the corresponding side of the other, respectively, then the triangles are congruent.

1. Work with your partner to explain how Angle-Angle-Side Congruence Theorem is true. Use the triangles shown to provide an example if desired.



Answers will vary. AAS Congruence Theorem must be true because any two triangles that have 2 congruent angles must also have the third angles congruent. Since we also have any one side congruent, we could prove congruence also using the ASA Congruence Theorem.

When introducing the Angle-Angle-Side (AAS) Congruence Theorem, tie it back to what students have explored in the prior activity so that they can connect the formal definition to their discovery from earlier.

Have students share their thinking to question 1. This will provide all students with an opportunity to hear other explanations from their peers, but also provide you with potential misconceptions that can be addressed.

2. Complete the statements.

If two angles of one triangle are congruent to two angles of another triangle, then the third pair of angles are [○ similar. ○ congruent.] If a congruent side opposite one of the given angles is congruent to the corresponding side of the other triangle, then [○ SSS ○ SAS ○ ASA] also applies. Thus, AAS is enough to show congruence.

The Hypotenuse-Leg (HL) Theorem

Two right triangles are congruent if the hypotenuse and a leg of one are congruent to the hypotenuse and a leg of the other, respectively.

3. The previous lesson's exploration showed that SSA is not sufficient to prove congruence. Explain why HL is sufficient to prove congruence for right triangles, despite being two sides with a non-included angle.
Answers will vary. HL is sufficient for right triangles because I can find the length of the other leg using the Pythagorean Theorem, confirming that the third side of both triangles are congruent.

When introducing the Hypotenuse-Leg (HL) Theorem, tie it back to what students have explored in the prior activity so that they can connect the formal definition with their discovery from earlier.

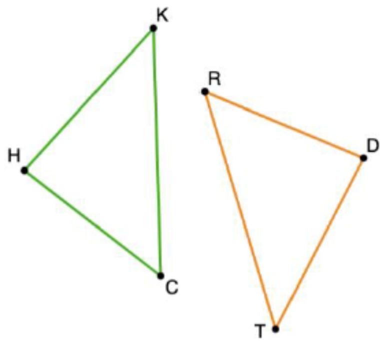
Would HL work in situations where the triangles are not right triangles? Why or why not?

5.3.5 Try It: Which Criteria?

Component Overview: *Students will practice determining if two triangles are congruent and which postulate or theorem would be used. This activity can be done independently to determine student understanding of triangle congruency criteria.*

5.3.5 Try It: Which Criteria?

1. Triangles HKC and DTR are shown. Determine if the given information is sufficient to determine the triangles are congruent. If so, name the congruence theorem or postulate that shows that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\angle KCH \cong \angle TRD$	No	
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\angle CHK \cong \angle RDT$	No	
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\overline{HC} \cong \overline{DR}$	Yes	AAS
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\overline{HK} \perp \overline{HC}$, $\overline{DT} \perp \overline{DR}$	Yes	HL
$\overline{HK} \cong \overline{DT}$, $\overline{HC} \cong \overline{DR}$, $\angle KHC \cong \angle TDR$	Yes	SAS

For this activity, begin by providing students with the direction. It may be necessary to remind them that this table includes all theorems and postulates they've learned thus far, including this lesson.

To differentiate the product, consider using the students' labeling of the given information and the naming of the parts on the triangles when determining the level of mastery of congruent triangles.

For a challenge, have students create triangles and/or conditions that would result in triangles being congruent under a theorem or postulate not used in the table.

5.3.6 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on their level of understanding for each congruence postulate or theorem. Students should complete this activity independently in order for the instructor to get an accurate reflection of student understanding.*

5.3.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each congruence postulate or theorem.

I need help. I got it. I could help someone.

Side-Side-Side (SSS)			
Side-Angle-Side (SAS)			
Angle-Side-Angle (ASA)			
Angle-Angle-Side (AAS)			
Hypotenuse-Leg (HL)			

Answers will vary.

This activity provides students with an opportunity to complete a self-reflection on their level of understanding of each congruence postulate or theorem they have learned so far in the unit. This data, coupled with unit performance, can provide valuable insight into which students may need additional support.

5.4 Exploring Corresponding Parts of Congruent Triangles Are Congruent

Lesson Introduction

Benchmarks	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. MA.912.GR.2.3 Specify a sequence of transformations that will map a given figure onto itself, or onto another congruent or similar figure. MA.912.GR.2.6 Apply rigid transformation to map one figure onto another to justify that the two figures are congruent.	Learning Targets	• I can relate the definition of congruence in terms of rigid motions to corresponding parts of congruent triangles. • I can use corresponding parts of congruent triangles to write congruence statements.

Benchmark Coherence	**Building On**	**Working Toward**
	MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.	N/A
Essential Questions	• How can I name corresponding parts when two triangles are congruent? • How can Corresponding Parts of Congruent Triangles are Congruent (CPCTC) be used to determine which parts of a triangle are congruent?	
Lesson Narrative	In this lesson, students will continue their learning of triangle congruence by extending the scenarios to include the use of rigid motions. Students will also learn the concept of CPCTC. They will practice writing congruence statements of corresponding parts from congruent triangles.	

Component Summary	5.4.1 Warm-Up (~5 min.)	Notice and wonder about geometry art *(SE p. 241)*	5.4.3 Guided Instruction (15-20 min.)	Formal instruction on CPCTC *(SE p. 243)*
	5.4.2 Collaboration (10-15 min.)	Collaboration on corresponding parts and rigid motions of triangles *(SE p. 242)*	5.4.4 Guided Practice (5-10 min.)	Practice opportunity naming corresponding parts of congruent triangles *(SE p. 245)*
	5.4.5 Wrap-Up (~5 min.)	Cool Down naming corresponding parts of congruent triangles and an Error Analysis *(SE p. 246)*		

Resources	**Glossary**	**Materials**	**Additional Preparation**
	<u>Key Terms</u>: • Definition of Congruence in Terms of Rigid Motions • Corresponding Parts of Congruent Triangles are Congruent (CPCTC) <u>Additional Terms</u>: N/A	• (Optional) Markers • (Optional) Patty paper	• This lesson guides students through a collaborative activity to review rigid motions and corresponding parts. Consider doing this activity on your own prior to the lesson to highlight potential misconceptions that may arise and to better understand where the activity is going.

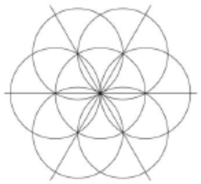
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may make mistakes when conducting the rigid motions.
Students may make errors when determining corresponding parts, particularly if the triangles are rotated.</td><td>Students should have a working knowledge of rigid motions and how they function. However, if your students struggle with them, consider doing a brief review of the different rigid motions and their properties.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
5.4.1 Warm-Up asks students to review patterns and designs. Students will describe what they notice about a design. Then, students will describe what they wonder about the pattern or final design when they review the last step of the 12-fold pattern.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will connect their prior learning of rigid motions and their properties to determine if two triangles are congruent.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider pairing students based on their performance in the prior set of lessons. Their understanding of what they have learned so far about triangle congruence can be a way to determine which students may need additional support.
Consider the use of physical manipulatives or patty paper during 5.4.3 Guided Instruction in order for visual learners to be able to picture which parts correspond with others.
5.4.5 Wrap-Up is a Cool Down that provides students with the opportunity to display mastery of the content from this lesson. Student response data can provide information on the degree to which students have mastered the content of this lesson. Consider having students mark the congruent parts on the figure as part of the data collected to measure mastery.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.4.1 Warm-Up: Art in Geometry

Component Overview: *Students will explore geometry that exists within art and designs. For this activity, students can work independently or with a small group. Consider having students, at a minimum, discuss the things they notice and things they wonder with a partner.*

5.4.1 Warm-Up: Art in Geometry

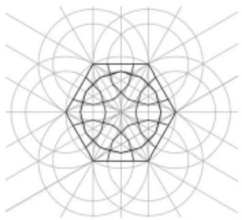
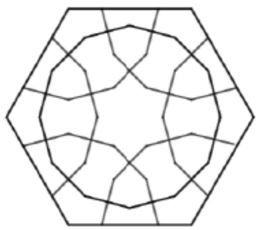
Salih is taking an art class that looks at the different patterns in tiles from across the world. Salih used a compass to create this first step in a 12-fold pattern.



For this activity, consider reading the narrative with the students out loud. It is important, for this context, for students to know that this is a 12-fold pattern prior to answering questions 1 and 2.

1. What do you notice about the design?
   Answers will vary. I notice that the design is very symmetrical. I also notice that the circles and lines all overlap.

The last step of the 12-fold pattern and the final design are shown.

Last Step	Final Design
	

Allow students to share out the things that they noticed and wondered. Consider listing things that students share in a chart so that students can hear alternative perspectives.

2. What do you wonder about the pattern or the final design?
   Answers will vary. I wonder how the artist was able to start with circles and end with a hexagon. I also wonder how the pattern repeats so perfectly.

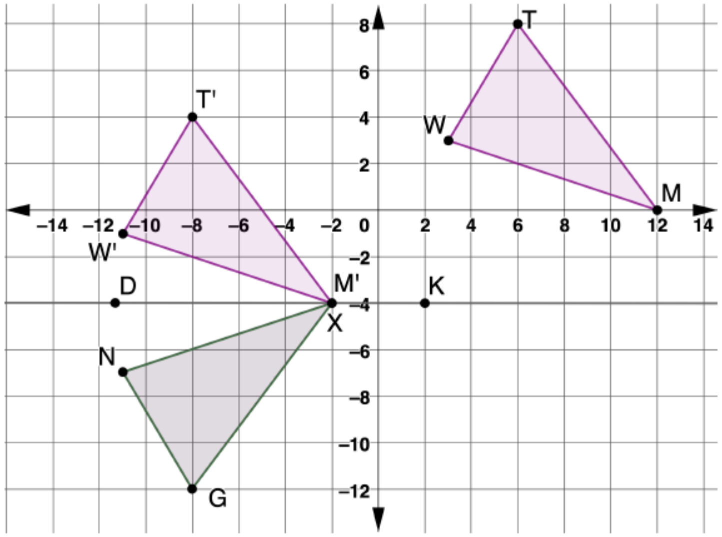
5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Component Overview: *Students will work together to discover corresponding parts and rigid motions of triangles. For this activity, release students to complete questions 1 and 2 with a partner and then come back together as a class to discuss. Review the formal definition for congruence in terms of rigid motion and then release students to continue working with their partners on questions 3 and 4.*

5.4.2 Collaboration: Congruence in Terms of Rigid Motions

Work with your partner on the following.

The diagram shows a sequence of transformations that, when applied to ΔTMW, show that ΔTMW maps to ΔGXN.



1. What are the transformations used to map ΔTMW to ΔGXN?
   The transformation was a slide 14 units to the left and 4 units down. Next the triangle was reflected over the line $\overleftrightarrow{DK}(y = -4)$

If students have struggled with rigid motions in the past, consider doing a brief review of basic rigid motions prior to launching this activity. If you do this, be cautious of your time so as to not make it a full lesson on rigid motions in isolation.

Notice and Wonder
What are some things that you notice about the triangles on the coordinate grid? What are some things that you wonder about them?

Some students may describe the transformations as two separate translations and a reflection instead of one translation and a reflection, as seen in the answer key. Both are correct and should be accepted. The student's response here determines how they will respond to question 4.

When reviewing the answers, consider having students come to the board or document camera to show the steps of the transformation for question 1 to occur, using a copy of ΔTMW.

5.4.2 Collaboration: Congruence in Terms of Rigid Motions (continued)

2. Complete the table by determining which parts of ΔGXN correspond with the given part of ΔTMW after mapping ΔTMW to ΔGXN.

ΔTMW	ΔGXN
$\overline{MT}$	$\overline{XG}$
$\overline{MW}$	$\overline{XN}$
$\overline{WT}$	$\overline{NG}$
$\angle TMW$	$\angle GXN$
$\angle TWM$	$\angle GNX$
$\angle WTM$	$\angle NGX$

Definition of Congruence in Terms of Rigid Motions

Two figures are congruent if and only if there exists one or more rigid motions, which will map one figure onto the other.

3. Discuss with your partner how to determine if the transformations that were used to map ΔTMW to ΔGXN are rigid motions.
Answers will vary. Yes, two transformations were used, a translation and a reflection. Both of the transformations are rigid motions because they preserve the distance and angle measure.

4. Complete the statement.

ΔTMW and ΔGXN are congruent because there exists

○ 1
○ 2
○ 3

rigid motion(s) which map(s) ΔTMW to ΔGXN.

Consider using colored markers or highlighters to draw attention to the line segments and angles in one triangle and matching the color to its corresponding part in the other triangle. Each part should be represented with a unique color so that students can connect the corresponding parts.

Spend some time reviewing the formal definition of congruence in terms of rigid motions. Don't just read the narration but connect the definition to the work they just did in questions 1 and 2. This formal definition has been informally explored since Unit 1. Address the "if and only if" part of the definition so the students know that both parts must be true. The foundation of this definition was laid in Unit 1, where students explored transformations as the plane moving and how any point on the preimage can be mapped to its corresponding point on the figure.

Why do you think mapping one figure to another shows congruence?

Some students may have seen the sequence of transformations in question 1 as two translations and a reflection. If this is the case, students will respond to question 4 as 3 rigid motions.

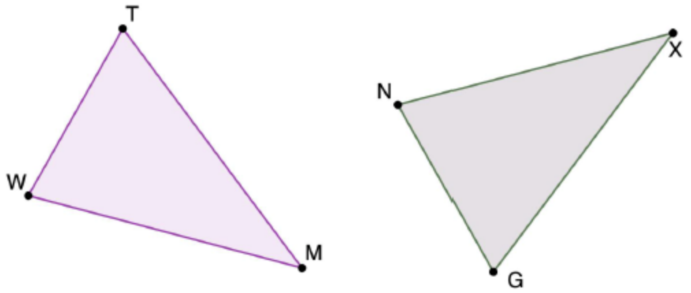
5.4.3 Guided Instruction: CPCTC

Component Overview: *Students will receive guided instruction on Corresponding Parts of Congruent Triangles are Congruent (CPCTC). For this activity, review the context of each problem with students prior to releasing them to try it on their own. Allow students to work with a partner where it's indicated.*

5.4.3 Guided Instruction: CPCTC

Another way to show that triangles are congruent is with congruence postulates and theorems.

1. ΔTMW and ΔGXN are shown where $\angle GNX \cong \angle MWT$, $\angle NXG \cong \angle TMW$, and $\overline{XG} \cong \overline{MT}$.



   a. From the given information, what congruence postulate or theorem can be used to show that $\Delta TMW \cong \Delta GXN$?
   Angle-Angle-Side

   b. Discuss with your partner what could be concluded about the corresponding parts of the triangles if the congruence postulate or theorem is used to show the triangles are congruent.

Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

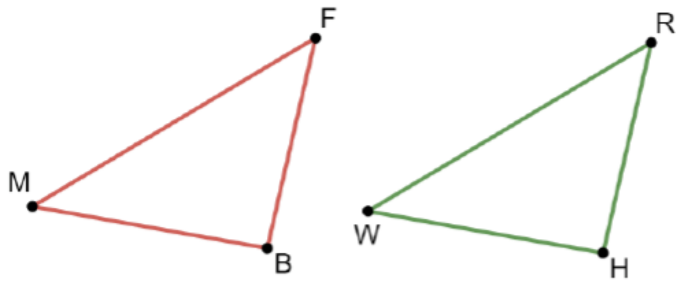
For visual learners, consider modeling how to mark up the triangle using geometric symbols and markings to translate the written information onto the visual of the triangles provided. There are many ways students can mark figures. Encourage students to develop their own strategy that they understand for marking congruent parts.

Consider leveraging a polling protocol where you can poll students on which postulate or theorem can be used to show that the triangles are congruent.
Have students defend their answer or explain their rationale and then do a re-poll to see if anyone was convinced based on the evidence.

5.4.3 Guided Instruction: CPCTC (continued)

There is a difference between naming two triangles and stating that two triangles are congruent. Order is important in a congruence statement, but is not necessary when only naming two triangles. The order used for congruence statements communicates which parts of the triangles correspond.

2. A diagram of two triangles is shown.



Naming	Congruence
ΔMFB and ΔHRW	$\Delta MFB \cong \Delta WRH$

Use the congruence statement to name all of the corresponding parts of the triangles.

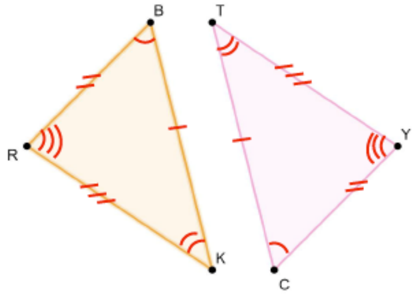
ΔMFB	ΔWRH
$\overline{MF}$	$\overline{WR}$
$\overline{FB}$	$\overline{RH}$
$\overline{MB}$	$\overline{WH}$
$\angle MFB$	$\angle WRH$
$\angle FBM$	$\angle RHW$
$\angle BMF$	$\angle HWR$

When reviewing the definition for Corresponding Parts of Congruent Triangles are Congruent (CPCTC), be sure to emphasize the importance of correctly identifying corresponding parts. Also, connect this concept back to what students did in prior problems.

Is there any way to verify that you are correct?

When reviewing question 2, consider using a document camera and a copy ΔMFB on patty paper to show the corresponding parts.

3. Consider triangles BKR and CTY.



a. If $\Delta BKR \cong \Delta CTY$, mark the corresponding congruent sides and the corresponding congruent angles on the diagram.

b. Complete each congruence statement.

$\overline{BK} \cong$ __CT___ $\angle KBR \cong \angle$__TCY_

__BR_____ $\cong \overline{CY}$ $\angle RKB \cong \angle$__YTC__

$\overline{RK} \cong$ __YT_____ $\angle$__BRK__ $\cong \angle CYT$

4. Consider two congruent triangles, ΔGRD and ΔFHW.

a. Write the congruence statement $\Delta GRD \cong \Delta FHW$ in two other ways.
$\Delta RGD \cong \Delta HFW$ $\Delta DRG \cong \Delta WHF$

b. Name the congruent sides for $\Delta GRD \cong \Delta FHW$.
$\overline{GR} \cong \overline{FH}$ $\overline{RD} \cong \overline{HW}$ $\overline{GD} \cong \overline{FW}$

c. Name the congruent angles for $\Delta GRD \cong \Delta FHW$.
$\angle GRD \cong \angle FHW$
$\angle RDG \cong \angle HWF$
$\angle DGR \cong \angle WFH$

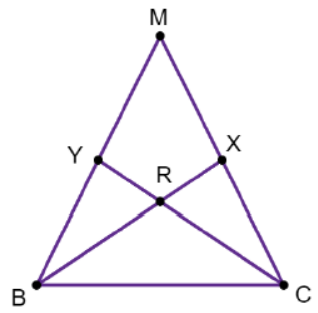
There are many ways that students can mark figures. Two examples are shown (using colors and using tick marks). There is not one "right" way. Rather, what is important is that students have a strategy they understand and use to mark figures after reading given information.
It's helpful to have highlighters on hand if students prefer to use color to mark corresponding parts.

What role does the labeling of the vertices play when determining which parts of a triangle are corresponding parts of the other? How can labeling them incorrectly impact the result?

Why do you think it is possible for triangle congruence statements to be written in multiple ways, yet still be true in terms of congruency?

5.4.3 Guided Instruction: CPCTC (continued)

5. Triangle MBC is shown.



a. How many different triangles are in ΔMBC?
8 triangles

b. If $\Delta BRY \cong \Delta CRX$, then complete the congruence statement.

$\overline{RC} \cong$ __$\overline{RB}$__ $\quad \angle RYB \cong$ _$\angle RXC$_

c. If $\Delta BXM \cong \Delta CYM$, then complete the congruence statement.

$\overline{CY} \cong$ __$\overline{BX}$__ $\quad \overline{CM} \cong$ __$\overline{BM}$__ $\quad \angle MCY \cong$ _$\angle MBX$_

Overlapping triangles can be difficult for some students. Students may not struggle with ΔBRY and ΔCRX, but may struggle with ΔBXM and ΔCYM. Have students draw ΔBRY and ΔCRX separately before marking and then repeat this by having them draw ΔBXM and ΔCYM separately before marking.

When reviewing question 5a, consider using different-colored highlighters to notate each unique triangle in the shape in order for students to better visualize the triangles that exist.

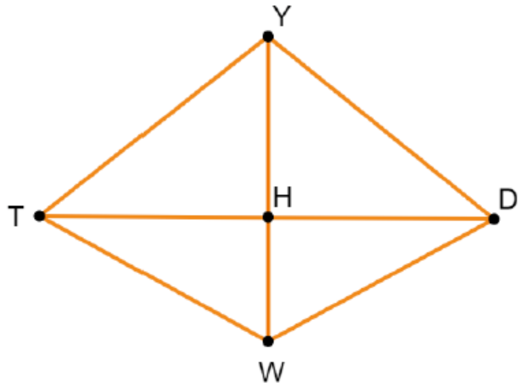
Why shouldn't $\angle RYB$ be simplified to just $\angle Y$ or $\angle RXC$ be simplified to just $\angle X$?

5.4.4 Guided Practice: Using CPCTC

Component Overview: *Students will engage in a practice opportunity on CPCTC. For this activity, review each question and allow students to try to answer independently. Then, review each answer prior to moving forward.*

5.4.4 Guided Practice: Using CPCTC

Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



1. How many different triangles are in Quadrilateral $YDWT$?
   8 triangles

2. If $\Delta DHY \cong \Delta THY$, then complete each congruence statement.

   $\overline{HD} \cong$ __$\overline{HT}$__ $\angle DYH \cong$ _$\angle TYH$_ $\overline{TH} \cong$ ___$\overline{DH}$__

3. If $\Delta DHW \cong \Delta THW$, then complete each congruence statement.

   $\angle TWH \cong$ _$\angle DWH$_ $\angle HTW \cong$ _$\angle HDW$_ $\overline{TW} \cong$ __$\overline{DW}$__

4. If $\Delta YTW \cong \Delta YDW$, then complete each congruence statement.

   $\angle WTY \cong$ _$\angle WDY$_ $\overline{DW} \cong$ __$\overline{TW}$__ $\overline{YW} \cong$ __$\overline{YW}$__

Students may struggle with seeing the individual triangles within the given quadrilateral for this activity.

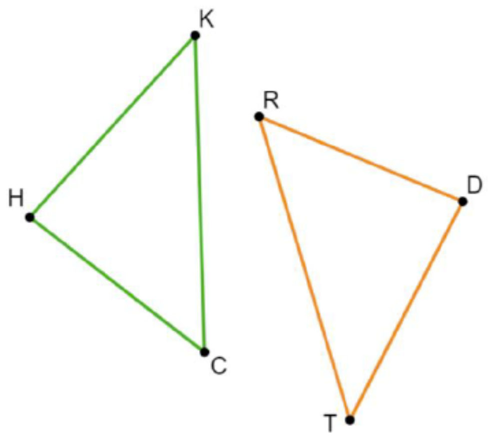
Students who struggle with questions 2-4 may have a misconception about determining corresponding parts of triangles. Review student answers to these questions to see if students may be misinterpreting the way the triangles are written in the congruency statement.

5.4.5 Wrap-Up: Cool Down

Component Overview: *Students will complete a Cool Down that assesses student understanding of the lesson content. This activity should be done independently to determine student mastery.*

5.4.5 Wrap-Up: Cool Down

The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$.



1. Complete each congruence statement.

   $\overline{KC} \cong$ _$\overline{TR}$__ $\overline{CH} \cong$ _$\overline{RD}$_

   $\angle RTD \cong$ _$\angle CKH$_ $\angle RDT \cong$ __$\angle CHK$_

2. Becky told her teacher that there was a typo in the first sentence. She circled the typo.

   The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$.

   Write a note to Becky explaining why this is NOT a typo.
   Answers will vary. ΔHKC and ΔTDR is naming the triangles, so order does not matter. The order of the vertices only matters when declaring congruence.

The data from the student response to this activity can provide insight on whether or not the students need additional support on the content from this lesson.
If students miss question 1, they likely misunderstand the connection between the congruence statements and their corresponding sides and angles. Encourage students to mark the congruent parts of the triangles for help in assessing whether students understand the concept of congruent parts but not the congruency statement.
If students miss question 2, they may have forgotten that in the naming of triangles, order does not matter.

5.5 Drawing Conclusions From Given Information

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg. MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.	Learning Targets	• I can draw and mark figures using given information. • I can use definitions, properties, postulates, and theorems to draw conclusions about triangles from about given information.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How can I use the information provided to mark figures appropriately? • How can I draw conclusions from given information?
Lesson Narrative	In this lesson, students will dive deeper into appropriately marking figures using given information. With the correct markings and other information, students will draw conclusions to solve problems.

Component Summary	5.5.1 Warm-Up (~5 min.)	Working through challenges *(SE p. 247)*	5.5.4 Collaboration (10 min.)	Collaboration on congruent overlapping triangles *(SE p. 251)*
	5.5.2 Collaboration (10 min.)	Collaboration on marking figures using given information *(SE p. 247)*	5.5.5 Try It (5 min.)	Practice drawing conclusions from given information *(SE p. 251)*
	5.5.3 Guided Instruction (10-15 min.)	Drawing conclusions from given information *(SE p. 249)*	5.5.6 Wrap-Up (~5 min.)	Ticket Out the Door on drawing conclusions from given information *(SE p. 252)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: N/A	• Colored Markers	• Work through each of the problems in advance to follow the flow of the lesson to help with facilitating the collaborative activities. • 5.5.1 Warm-Up, contains a video to accompany the question prompts

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may misread the labeling and make incorrect markings.
Students may not make their markings unique, leading to confusing information.
Students may not struggle with overlapping triangles.</td><td>In this lesson, students are drawing information from given shapes. Students may struggle with reading and interpreting the information correctly. Consider using colored highlighters or markers to help link each identified piece of information that can be connected between the image and the statements.
If time is an issue, consider assigning 5.5.5 Try It for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
5.5.2 and 5.5.4 Collaborations provide opportunities for students to work together. Students will be able to try the problems on their own and then pair up with their partner to share their work with one another and come to a consensus.</td><td>MA.K12.MTR.5.1: Patterns and Structure
Students will be drawing conclusions from given information throughout this lesson. Students will need to make appropriate markings and complete various components of the lesson to solve more complex problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider pairing students based on their performance in the prior set of lessons. Student's understanding of what they have learned so far about triangle congruence can be a way to determine which students may need additional support.
Consider the use of physical manipulatives during 5.5.4 Collaboration for visual learners to be able to visualize the various components when working with overlapping triangles.
5.5.6 Wrap-Up includes a Ticket Out the Door that provides students the opportunity to display mastery of the content from this lesson. Student response data can provide information on the degree to which students have mastered the content of this lesson. Consider evaluating each part of the question separately when collecting data to evaluate understanding.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.5.1 Warm-Up: Working Through Challenges

Component Overview: *Students engage in an activity about working through challenges. For this activity, show the video for a more in-depth conversation during the interview with the student, Teri.*

5.5.1 Warm-Up: Working Through Challenges

Teri takes dance in school, but learning how to tap dance was a new and overwhelming experience. She said that after her teacher introduced tap, "It was really hard for me to learn ... It just looked like a lot of work and looked like it took a lot of time to learn, and it was really high tempo."

1. What is something that felt overwhelming the first time you were learning it?
   Answers will vary. I found learning a foreign language to be overwhelming the first time I tried to learn it.

Teri's teacher slowed down the moves to show her the structure and recommended that she practice each step before trying to put it all together. She connected this experience to learning other things that are difficult, saying, "The more you practice a certain skill that you need, the better you become at it, and the easier it becomes."

2. Describe a time when practicing something that was challenging for you helped you become better at it.
   Answers will vary. I practiced a musical instrument and over time was able to get much better at it. I practiced a sport and over time was able to get much better at it. I practiced a video game and over time was able to get much better at it.

For this activity, consider modifying this to be a think-pair-share activity. Since this does not require any specific mathematical knowledge, students who may not often participate will have the same entry point as others and be able to engage in the conversation.

Consider having students pair up with a partner to share their answers to questions 1 and 2. Allow each partner to share their partner's story to improve listening and communication skills in conjunction with the discussion on working through challenges.

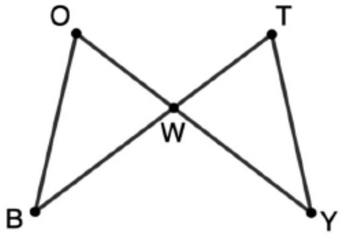
5.5.2 Collaboration: Using Given Information to Mark Figures

Component Overview: *Students will work together to determine how to accurately mark figures using given information. For this activity, have students work with a partner to complete question 1 and then review answers together as a class. Then, release students to complete questions 2 and 3.*

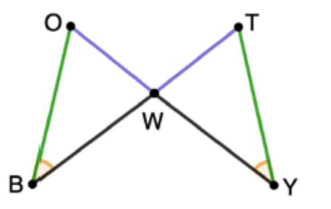
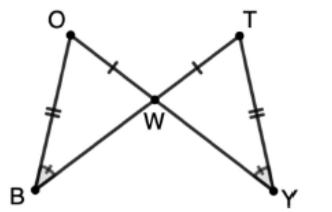
5.5.2 Collaboration: Using Given Information to Mark Figures

When given information about geometric figures, it is helpful to mark the figures.

1. Triangles OBW and WTY are shown.



Josue and Abby marked the figure in different ways given $\overline{OW} \cong \overline{TW}$, $\overline{OB} \cong \overline{TY}$, and $\angle OBW \cong \angle TYW$ as shown.

Josue's Markings	Abby's Markings
	

a. Discuss with your partner what you notice about how Josue and Abby each marked the figure.

b. Complete the table to describe similarities and differences in the way each student marked the figure.

How were their markings similar?	How were their markings different?
Both methods clearly show congruent angles and sides.	Josue used different colors to identify congruent sides and angles. Abby used tick marks to identify congruent sides and angles.

Notice and Wonder

- *What are some things that you notice about the triangles shown in question 1?*
- *What are some things that you wonder about the triangles shown in question 1?*

There are many ways that students can mark figures. Two examples are shown (using colors and using tick marks). There is not one "right" way. Rather, it is important that students have a strategy they understand and use to mark figures after reading given information.

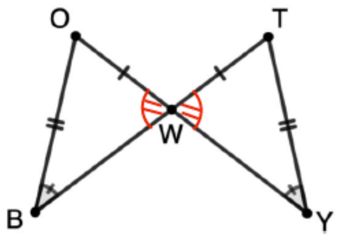
It's helpful to have highlighters on hand if students prefer to use color to mark corresponding parts.

When reviewing answers, have students share their responses to both the similarities and the differences. Consider displaying answers that are being shared. Help students to realize that the method used for marking is not as important as the accuracy of the markings themselves. Allow students to know that they can choose a method that best works for them.

c. Josue wondered if there was enough information given to conclude $\Delta OWB \cong \Delta TWY$. Work with your partner to explain whether there is enough information to conclude $\Delta OWB \cong \Delta TWY$.
Answers will vary. No there is not enough information. $\angle WOB$ can differ from $\angle WTY$, and this would change the length of sides $\overline{WB}$ and $\overline{WY}$ respectively. Therefore, SSA is not able to prove triangles congruent.

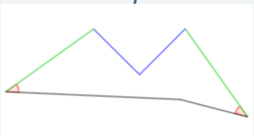
d. Abby said she does not think there is enough given information to conclude $\Delta OWB \cong \Delta TWY$, but she said there is enough information to conclude $\angle OWB \cong \angle TWY$. What definition is Abby using to draw this conclusion?
Abby used the Vertical Angles Theorem.

e. Add markings to Abby's figure to represent her conclusion that $\angle OWB \cong \angle TWY$.



f. Work with your partner to explain whether there is now enough information to conclude $\Delta OWB \cong \Delta TWY$.
Answers will vary. Yes, now that I know two angles and at least one side congruent, there is enough information to conclude congruence with Angle-Angle-Side.

On question 1c, to provide a visual understanding of why not enough information was given, draw the figure with the congruent parts and with the other parts purposefully not congruent.
For example:

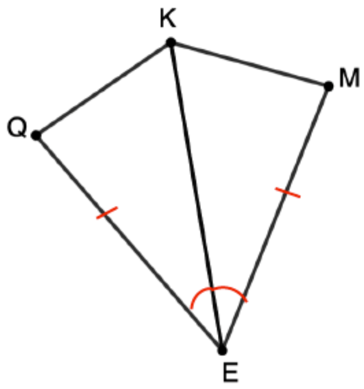


- *How did you determine which markings could be added to the figure?*
- *What could be used to justify that your markings are correct?*

Consider splitting the class into four groups, each with a different congruency postulate or theorem. Then have each group either prove why the triangles could be proven congruent or why they could not be proven congruent, using their assigned postulate or theorem.

5.5.2 Collaboration: Using Given Information to Mark Figures (continued)

2. Mark polygon $KQEM$ using the given information.
$\overline{EQ} \cong \overline{EM}$ and $\angle KEQ \cong \angle KEM$



Students could use one of the strategies modeled from Josue's or Abby's work, or they could use another system of marking that works for them. Keep an eye out for ways that students mark their figures and have students share their strategies if any different ones are used.

3. Compare your markings with your partner. Make any corrections if necessary.

5.5.3 Guided Instruction: Drawing Conclusions From Given Information

Component Overview: *Students will receive guided instruction on drawing conclusions from given information. For this activity, guide students through each question in the activity.*

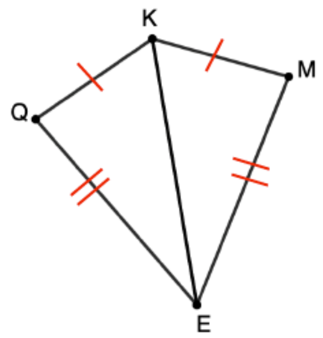
5.5.3 Guided Instruction: Drawing Conclusions from Given Information

Given information, definitions, properties, postulates, and theorems can be used to draw conclusions about figures.

1. What are some definitions, properties, postulates, and theorems related to triangles and angles learned so far this year?

Definitions	*Answers will vary. midpoint, perpendicular lines, perpendicular bisector, right triangles, angle bisector*
Properties	*Answers will vary. Reflexive property of congruence, transitive property, all angles of a triangle add up to* $180°$.
Theorems and Postulates	*Answers will vary. SSS, SAS, ASA, AAS, HL, Vertical Angles Theorem, Corresponding Angles Theorem*

Polygon $KQEM$, where $\overline{QK} \cong \overline{KM}$ and $\overline{EQ} \cong \overline{EM}$ is shown. $\overline{KE}$ is drawn inside polygon $KQEM$.



2. Mark the figure using the given information.

3. Discuss any conclusions that can be drawn about the figure with the information given.

Question 1 is not meant to be an exhaustive list; rather, an opportunity to brainstorm concepts that may come into play when drawing conclusions. This should also not take a lot of time but provide a quick primer for the remainder of the lesson. Students can add to this table as additional definitions, properties, postulates, and theorems are used throughout the lesson.

Consider using question 1 as a whole class share-out activity. For example, an entry point that is accessible to all is a silent board activity. Put the three categories on the whiteboard and have students come up silently to add an item not already listed as they think of each new item. Students can choose which category to write in. You can also create a game out of this by creating two teams and having students add items one at a time until the team can no longer add anymore, giving the team a point for being the last one to add a definition, property, or theorem and postulate.

Give a little extra time for students to share their markings and conclusions in questions 2 and 3 with all the students in the class. Getting students to understand these two elements is critical to learning the concepts of this unit.

5.5.3 Guided Instruction: Drawing Conclusions From Given Information (continued)

Using the given information, Hilary concludes $\Delta QEK \cong \Delta MEK$.

4. Complete the statement to explain how Hilary is able to draw this conclusion.

   Hilary noticed
   - ○ 1
   - ○ 2
   - ○ 3

   corresponding
   - ○ side(s)
   - ○ angle(s)

   of ΔQEK and ΔMEK are congruent based on the given information.

   Both triangles also share a common
   - ○ side,
   - ○ angle,

   which is congruent to itself because of the
   - ○ reflexive
   - ○ transitive

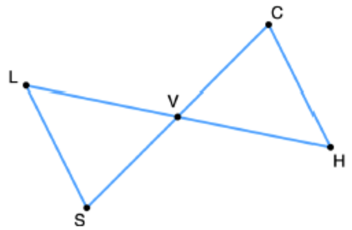
   property.

   She can conclude $\Delta QEK \cong \Delta MEK$ because of
   - ○ SSS.
   - ○ SAS.
   - ○ ASA.
   - ○ AAS.
   - ○ HL.

To set students up for success with what they'll be asked to do, consider asking the following questions:
- *What conclusion(s) did Hilary draw?*
  - $KE = KE$
  - *the triangles are congruent*
- *Why was she able to draw those conclusions?*
  - *reflexive property*
  - *SSS*

5. Triangle LVS and triangle CVH are shown.

   $\overline{SV} \cong \overline{VC}$ and
   $\angle LSV \cong \angle HCV$



   a. What conclusions can be drawn from the given information?
   Answers will vary.
   Triangles VLS and VHC are congruent since $\angle LVS$ and $\angle HVC$ are vertical angles and vertical angles are congruent by the Vertical Angles Theorem.

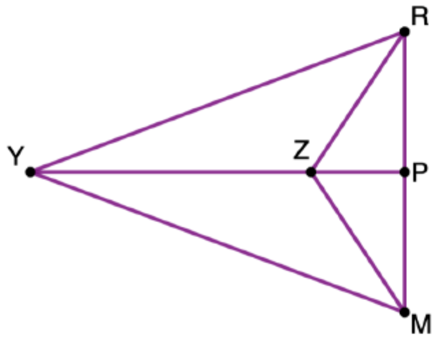
Allow students to share the conclusions they developed in question 5a. Consider posting various conclusions on a piece of chart paper or on a section of the whiteboard. Allow students to debunk any conclusions based on evidence.

What knowledge was necessary to make additional appropriate markings?

b. What congruence postulate or theorem is the reason these conclusions can be made?
Angle-Side-Angle

c. Julia says she can conclude $\overline{CH} \cong \overline{SL}$ and $\angle SLV \cong \angle CHV$. Explain why Julia is able to draw these conclusions.
Answers will vary. These conclusions can be drawn because once triangles are proven congruent, all corresponding parts will be congruent.

6. Triangle RMY is shown. $\angle YMZ \cong \angle YRZ$ and $\angle RZY \cong \angle MZY$.



Work with your partner to complete the statements based on the given information.

a. I can conclude ...
$\Delta YRZ \cong \Delta YMZ$

b. I know this because ...
By the reflexive property of congruence $\overline{ZY} \cong \overline{ZY}$ *so* $\Delta YRZ \cong \Delta YMZ$ *by AAS.*

Notice and Wonder
- *What do you notice about the triangle in question 6?*
- *What do you wonder about it?*
- *How would you go about marking this triangle?*

Have students compare their answers in question 6a and 6b with a partner. Then have each partner pair compare their work with another partner pair.

5.5.4 Collaboration: Overlapping Triangles

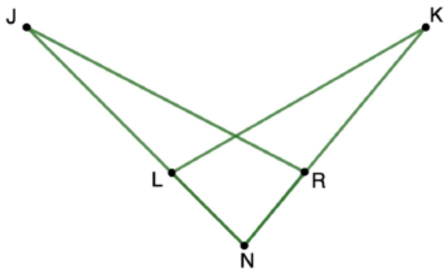
Component Overview: *Students will work together on problems that involve overlapping triangles. For this activity, allow students to work on both questions with a partner.*

5.5.4 Collaboration: Overlapping Triangles

Work with your partner to complete the following.

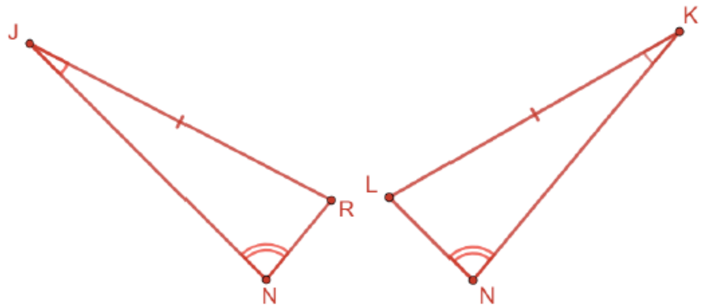
Dalton saw the following question with the polygon shown in an old geometry textbook.

"If $\angle NKL \cong \angle NJR$ and $\overline{KL} \cong \overline{JR}$, what else do you know so that you can show that $\Delta RJN \cong \Delta LKN$ using ASA?"



He knew that both triangles share $\angle N$. Therefore, by knowing that $\angle N \cong \angle N$ by the reflexive property of congruence, he concluded that he could use ASA to show that $\Delta RJN \cong \Delta LKN$.

1. Draw the triangles so that they do not overlap. Then, mark the triangles with the information given from the textbook.



When reviewing the answers for this activity, begin by reviewing the context and image provided at the beginning of this activity. Provide students with a think aloud about things you notice about the information.

Consider using patty paper to provide a visual of how to properly take the image of the two overlapping triangles and create two non-overlapping triangles that maintain their original properties, features, and labeling.

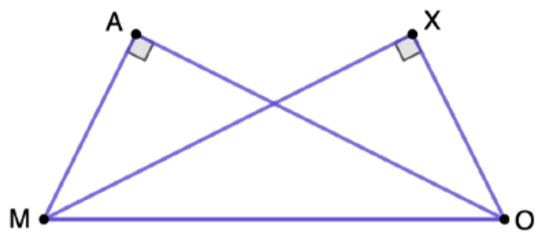
2. Explain why Dalton can or cannot use ASA to show that $\Delta RJN \cong \Delta LKN$.
Answers will vary. Although ASA is not the most direct route, he could use ASA. Using the triangle sum theorem, he knows the third angles will also be congruent, $\angle JRN \cong \angle KLN$. Once he does this, then he could use ASA.

5.5.5: Try It: Drawing Conclusions From Given Information

Component Overview: *Students will practice drawing conclusions from given information. For this activity, allow students to try this on their own to determine who may need additional support.*

5.5.5 Try It: Drawing Conclusions from Given Information

1. Triangle MAO and triangle XOM are shown, where $\angle MAO$ and $\angle OXM$ are right angles and $\overline{MA} \cong \overline{XO}$.



Complete each statement based on the given information.

a. I can conclude that ...
Answers will vary.
$\Delta MAO \cong \Delta OXM$

b. I know this because ...
Answers will vary.
I can use the reflexive property of congruence to declare that $\overline{MO} \cong \overline{MO}$. Using this and the fact that these are both right triangles allows me to use the HL Theorem to prove the triangles are congruent.

Consider using this activity as a time to work with small group of students who need additional support based on their performance in the prior activities.

Have students compare their answers in question 1a and 1b with a partner. Then have each partner pair compare their work with another partner pair.

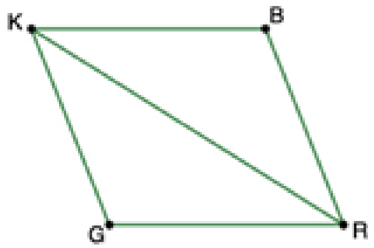
5.5.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students will complete a Wrap-Up that allows them to practice completing statements based on the given information. This activity is designed to be done independently to allow for an individual check for understanding on the content of this lesson.*

5.5.6 Wrap-Up: Ticket Out the Door

Quadrilateral $BRGK$ is shown.

Complete each statement based on the given information.



1. Given that $\overline{KB} \cong \overline{GR}$ and $\overline{KG} \cong \overline{BR}$,
   a. I can conclude that ...
      $\Delta KRG \cong \Delta RKB$
   b. I know this because ...
      I can use the reflexive property of congruence to know that $\overline{KR} \cong \overline{KR}$. This allows me now to use Side-Side-Side to know triangle KRG is congruent to triangle RKB.

2. Given that $\overline{KB} \cong \overline{KG}$ and $\overline{RG} \cong \overline{BR}$,
   a. I can conclude that ...
      $\Delta KRG \cong \Delta RKB$
   b. I know this because ...
      I can use the reflexive property of congruence to know that $\overline{KR} \cong \overline{KR}$. This allows me now to use Side-Side-Side to know triangle RKG is congruent to triangle RKB.

3. Given that $\overline{KG} \parallel \overline{BR}$ and $\overline{KB} \parallel \overline{GR}$,
   a. I can conclude that ...
      $\Delta KRG \cong \Delta RKB$
   b. I know this because ...
      When parallel lines are intersected by a transversal, alternate interior angles are congruent. Therefore, $\angle GKR \cong \angle BRK$ and $\angle GRK \cong \angle BKR$. By reflexive property, $\overline{KR} \cong \overline{KR}$ and allowing me to use Angle-Side-Angle.

The data from the student response to this activity can provide insight on whether or not the students need additional support with the content from this lesson. Questions 1 and 2 assess whether students recall the reflexive property and the conditions for SSS. Question 3 requires the student to recall alternate interior angles of parallel lines, the reflexive property, and the conditions for ASA.

5.6 Proving Triangles Congruent

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg. MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.	Learning Target	• I can prove two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can I prove that two triangles are congruent? • How can I determine which postulate or theorem best proves if two triangles are congruent?
Lesson Narrative	In this lesson, students extend their knowledge of triangle congruency to prove that two triangles are congruent. Students will complete two-column, flow, and paragraph style proofs in order to apply their knowledge from this unit.

Component Summary				
	5.6.1 Warm-Up (~5 min.)	Real-world estimation problem *(SE p. 253)*	5.6.3 Guided Practice (10 min.)	Practice with proving triangles congruent *(SE p. 255)*
	5.6.2 Guided Instruction (15 min.)	Formal instruction on proving triangles congruent *(SE p. 253)*	5.6.4 Your Turn! (10 min.)	Independent practice proving two triangles are congruent *(SE p. 256)*
	5.6.5 Wrap-Up (~5 min)	Error analysis of a flowchart proof with proving two triangles are congruent *(SE p. 256)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• (Optional) Colored Markers	• Students may struggle with the proofs in this lesson. Work through each one on your own prior to delivering instruction to work through potential misconceptions in the proofs. • Some proofs have alternate paths. Prepare knowing that students may consider these alternate paths.

Teacher Tips	**Common Misconceptions** • Students may not know the definitions used in the proofs. • Students may not realize all the information that is available in the image of a problem when it's not explicitly stated. • Student may apply the same marking to multiple sides, causing confusion.	**Things to Consider** • Consider asking students to complete a chart, notebook, or foldable for the definitions, postulates, and theorems that will be used in proofs. This can include common reasons that usually follow each other. For example, "definition of perpendicular lines" is typically followed by "definition of right angles" or "right angles congruence theorem." Also, the definition of a bisector is typically followed by a congruence statement.
	Literacy and Language Routines **Think-Pair-Share** 5.6.3 Guided Practice has students completing a flow chart proof to prove two triangles are congruent. For this activity, students can think through their own steps and reasoning and then pair up with a partner to share their answers together to come to a consensus.	**Mathematical Thinking and Reasoning (MTR) Standard** **MA.K12.MTR.5.1: Patterns and Structure** In 5.6.2 Guided Instruction, students will draw conclusions based on given information and explain why the conclusion can be drawn. Students will then order the steps to formally prove the conclusion from the given information.
Differentiate Instruction	Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it. For 5.6.2 Guided Instruction, consider using auditory explanations for a proof prior to filling it in. Use common language to describe how the triangles could be proven congruent. Then, transition that language into formal geometric language when completing the proof. The 5.6.5 Wrap-Up activity provides students with an opportunity to demonstrate their understanding of triangle congruence using a proof. The data provided from this activity can help determine which students may need additional support.	
Lesson Notes:		

5.6.1 Warm-Up: Cheese Balls

Component Overview: *Students will engage in a real-world estimation problem. Students may need to be probed on what shapes they see in the figure and what measurements will be used.*

5.6.1 Warm-Up: Cheese Balls

The image shows cheese balls arranged on a baking sheet with a white plate on top. There are thirteen cheese balls on the plate.

1. Estimate how many cheese balls will fit on the plate.
   Answers will vary. 130 *cheese balls.*

2. Explain how you determined your estimate.
   Answers will vary. If we estimate that the plate has a diameter of 13 *cheese balls wide, we could estimate the area of the plate as* $A = \pi(6.5^2) \approx 133$ *cheese balls.*

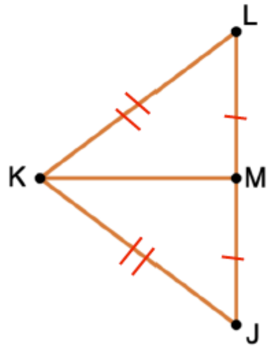
If you didn't want to use an estimation strategy and are asked for a specific answer, what information would you need to know in order to answer the question correctly? What would you do with the information to solve?

5.6.2 Guided Instruction: Proving Triangles Congruent

Component Overview: *Students will receive guided instruction on proving triangles are congruent through a flow chart proof, two-column proof, and paragraph proof.*

5.6.2 Guided Instruction: Proving Triangles Congruent

When completing a proof to show two triangles are congruent, it is helpful to start by considering conclusions that can be drawn from the given information.



1. Triangle KLJ is shown with some given information.

   Given: $\overline{KM}$ bisects $\overline{LJ}$
   $\overline{KL} \cong \overline{KJ}$

   a. What conclusion can be drawn from the given statement, "$\overline{KM}$ bisects $\overline{LJ}$"?

   $\overline{JM} \cong \overline{LM}$

   b. Explain why this conclusion can be drawn.
   Answers will vary. Because $\overline{KM}$ bisects $\overline{LJ}$, by the definition of segment bisector $\overline{LJ}$ is divided into two equal parts.

   c. Mark the figure based on the given information.
   Marked on diagram.

   d. What other conclusion(s) can be drawn from the information given?
   $\overline{KM} \cong \overline{KM}$

   e. Explain why this conclusion can be drawn.
   Answers will vary. By the reflexive property of congruence, $\overline{KM} \cong \overline{KM}$.

Call attention to the term "bisect." Students may generally understand the term but may forget what occurs when one segment bisects another. Then explain that the definition of a segment bisector or angle bisector results in equality of the bisected parts. To show that these parts are congruent, the definition of congruence should be used.

Consider having students share their rationale from question 1b. If available, use manipulatives to demonstrate the concept of bisecting again and connect that concept to the triangle from this question.

Keep in mind that students may choose to use alternative ways of marking than what is referenced in the answer key here.

Students may be need additional support in completing the proof for this question. Consider walking students through reviewing the conclusions that they have drawn from the question. Assist students with transitioning their logic into a flow proof. Draw attention to what is concluded after given a statement about a bisector. Students may notice that ΔKLJ is an isosceles triangle, concluding that $\angle L \cong \angle J$. This is an alternate way for students to prove that $\Delta KMJ \cong \Delta KML$. The flow chart proof in question 1f is laid out for SSS.

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

f. Use the conclusions drawn about ΔKLJ from the information given, along with the figure, to complete the flow chart proof.

Given: $\overline{KM}$ bisects $\overline{LJ}$

$\overline{KL} \cong \overline{KJ}$

Prove: $\Delta KMJ \cong \Delta KML$

$\boxed{\overline{KM} \text{ bisects } \overline{LJ}}$ → $\boxed{LM = MJ}$ → $\boxed{\overline{LM} \cong \overline{MJ}}$

Given | Definition of segment bisector | Definition of congruence

$\boxed{\overline{KM} \cong \overline{KM}}$

Reflexive Property of Congruence

$\boxed{\overline{KL} \cong \overline{KJ}}$

Given

$\boxed{\Delta KMJ \cong \Delta KML}$

SSS congruence

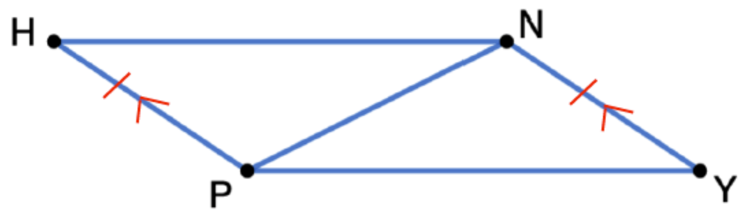
When reviewing the flowchart proof, repeat the rationale on why each of the steps is being identified in the process. Auditory learners can benefit from hearing each step along with the rationale for why those reasons are provided.

Does the order in which the boxes are completed matter? Sample answer: No. The only time order matters is if the box or boxes has an individual conclusion box following, as the first box in the first column does. The second and third boxes in the first column do not lead to an individual conclusion box — only the final conclusion to prove. Therefore, the second and third boxes in the first column could be swapped.

How could we prove the triangles are congruent using SAS?

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

2. Polygon $HPYN$ is shown.



a. Mark the diagram to show the following given information.

Given: $\overline{HP} \cong \overline{YN}$ and $\overline{HP} \parallel \overline{YN}$
Marked on diagram.

b. Discuss how you determined which markings to make with your partner.
Answers will vary. We selected markings that we could agree on.

c. Which congruence theorem or postulate should be used to prove $\Delta PHN \cong \Delta NYP$?
Side-Angle-Side

d. Complete the two-column proof.

Statement	Reason
1. $\overline{HP} \cong \overline{YN}$	1. Given
2. $\overline{HP} \parallel \overline{YN}$	2. *Given*
3. $\angle HPN \cong \angle YNP$	3. *Alternate Interior Angles Postulate*
4. $\overline{PN} \cong \overline{PN}$	4. *Reflexive Property of Congruence*
5. $\Delta PHN \cong \Delta NYP$	5. *SAS Congruence Postulate*

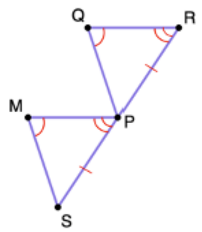
When reviewing the answers to questions 2a and 2b, consider having students come to the board to make a notation and explain their rationale. Continue this process until all markings have been addressed.

For question 2c, are there any additional congruence theorems or postulates that could be used in this example? Why or why not?

For question 2d, consider using a silent board strategy in which students silently go to the board and fill in one missing statement or reason that they know. This strategy provides an entry point for the students who only know some of the responses, but not all.

5.6.2 Guided Instruction: Proving Triangles Congruent (continued)

3. Triangles MSP and QPR are shown.



Given: $\overline{QR} \parallel \overline{MP}$
$\angle SMP \cong \angle PQR$
P is the midpoint of $\overline{SR}$

Prove: $\Delta SPM \cong \Delta PRQ$

a. Mark the diagram to show the given information.
Marked on diagram.

b. Complete the paragraph proof.

It is given that P is the midpoint of $\overline{SR}$, so __$\overline{SP} \cong \overline{PR}$_

by _*definition of midpoint* __. It is given that $\overline{QR} \parallel \overline{MP}$,

so _$\angle QRP \cong \angle MPS$__ by _ *Corresponding Angles Theorem*_.

It is also given that __$\angle SMP \cong \angle PQR$ _. Therefore,

$\Delta SPM \cong \Delta PRQ$ by _*AAS Congruence Postulate*_.

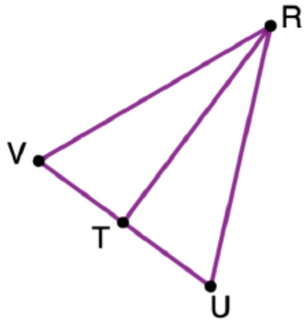
If students need additional support, consider creating a word bank for students to use in order to complete the paragraph proof. If you employ this strategy, create additional words that would not be correct for this situation.

5.6.3 Guided Practice: Proving Triangles Congruent

Component Overview: *Students will engage in guided practice with proving triangles are congruent. For this question, review the triangle and given information and allow students to try this on their own. Bring the class back together to review the answers.*

5.6.3 Guided Practice: Proving Triangles Congruent

1. Triangle VRU is shown.

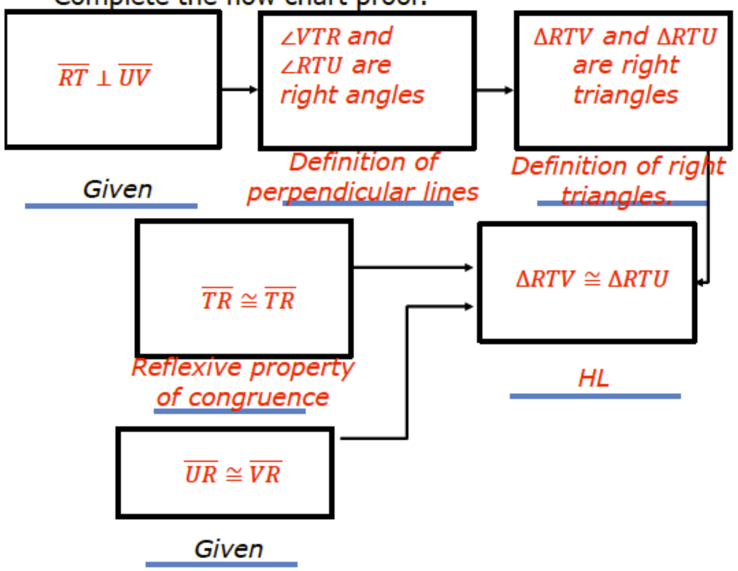


Given: $\overline{UR} \cong \overline{VR}$

$\overline{RT} \perp \overline{UV}$

Prove: $\Delta RTV \cong \Delta RTU$

Complete the flow chart proof.



Think-Pair-Share

For this activity, students can think through their own steps and reasoning and then pair up with a partner to share their answers together to come to a consensus.

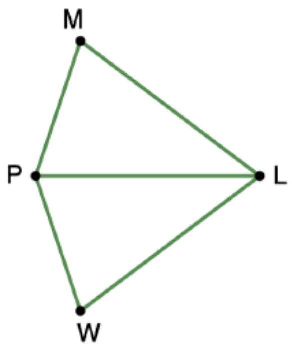
For students who finish early, consider having students take the information from a flow chart proof and write it into a paragraph proof for additional practice.

5.6.4 Your Turn!

Component Overview: *Students will practice proving two triangles are congruent on their own. For this activity, release students to complete it on their own.*

5.6.4 Your Turn!

1. Polygon $MLWP$ is shown.



Given: $\overline{LP}$ bisects $\angle MLW$
$\angle LPM \cong \angle LPW$

Prove: $\Delta WPL \cong \Delta MPL$

Complete the paragraph proof.

It is given that $\overline{LP}$ bisects $\angle MLW$, so $m\angle MLP = m\angle WLP$ by

definition of ___ *angle bisector* ________ therefore

____$\angle MLP \cong \angle WLP$ _ by definition of congruence. Also, it

is given that _____$\angle LPM \cong \angle LPW$ ___.

_PL____ $\cong$ _PL___ by _*reflexive property of congruence*_.

Therefore, $\Delta WPL \cong \Delta MPL$ by *ASA congruence*.

When reviewing the answer to this question, consider having students do a relay-style review. The relay could be done in the following order:

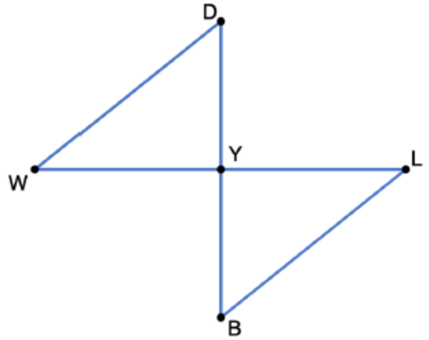
- *Student 1 – Mark the image*
- *Student 2 – Complete first fill-in-the blank of the paragraph proof*
- *Student 3 – Complete second fill-in-the blank of the paragraph proof, reading from the beginning through their answer*
- *Student 4 – Complete third fill-in-the-blank of the paragraph proof, reading from the beginning through their answer*
- *Student 5 – Complete fourth fill-in-the-blank of the paragraph proof, reading from the beginning through their answer*
- *Student 6 – Complete last fill-in-the-blank of the paragraph proof, reading from the beginning through their answer*
- *Student 7 – Read entire proof and defend/ illustrate with the shape markings*

5.6.5 Wrap-Up: Error Analysis

Component Overview: *Students will complete an error analysis of a flow chart proof of proving two triangles are congruent. This activity should be done independently to better assess each student's level of understanding of the content.*

5.6.5 Wrap-Up: Error Analysis

1. The flow chart proof has two errors. Find the errors and correct them.



Given: Y is the midpoint of $\overline{WL}$
$\overline{DY} \cong \overline{YB}$

Prove: $\Delta DWY \cong \Delta BLY$

Statement	Reason
Y is the midpoint of $\overline{WL}$	Given
$\cancel{\overline{DY} \cong \overline{BY}}$ $\overline{WY} \cong \overline{YL}$	Definition of a midpoint
$\overline{DY} \cong \overline{YB}$	Given
$\angle DYW \cong \angle BYL$	Vertical Angles Theorem
$\Delta DWY \cong \Delta BLY$	~~ASA~~ *SAS* congruence

This question assesses student understanding on proving two triangles are congruent. There are two potential scenarios that may arise from this data:

- *Students do not have mastery of triangle congruency*
- *Students do not have mastery of completing proofs*

Use student performance in this activity along with their performance in prior lessons to help determine which area students fall in and where additional support may be needed.

5.H1 Definition of Congruence in Terms of Rigid Transformations

Lesson Introduction

Benchmark	MA.912.GR.2.7 Justify the criteria for triangle congruence using the definition of congruence in terms of rigid transformations.	Learning Target	• I can use the definition of congruence to justify that SSS congruence, SAS congruence, ASA congruence, and AAS congruence are sufficient to show that two triangles are congruent.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can you use the definition of congruence to justify that SSS, SAS, ASA, and AAS congruence are sufficient to show that two triangles are congruent?
Lesson Narrative	In this lesson, students will extend their learning of congruence in terms of rigid transformations. Students will learn how to justify congruence through a structured exploration using various rigid transformations. Through the process of the different transformations, students will explore the connection between their patty paper verification and a series of transformations to understand that the definition of congruence in terms of rigid transformations shows that SSS, SAS, ASA, and AAS congruence are sufficient.

Component Summary	5.H1.1 Warm-Up (~5 min.)	Geometry as an animal activity *(SE p.* 257)	5.H1.2 Exploration (30-35 min.)	Justifying congruence activity *(SE p.* 258)
	5.H1.3 Wrap-Up (~5-10 min.)	Square, triangle, circle reflection activity *(SE p.* 261)		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• Patty Paper	• This lesson is designed for students enrolled in honors Geometry as the benchmark addressed is only included in the Geometry Honors course.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

- Students may incorrectly state the transformations that map a triangle onto another.
- Students may not connect that the given information is shown to be true by showing that the definition of congruence was applied by mapping the triangles.
- Students may conclude that two triangles are congruent without proper evidence.

</td><td>

- The primary function of this lesson is intended to be an exploration through Activity 5.H1.2. It is not a lesson to give students information, but rather to have them explore the concept to draw the conclusion on their own through the activity. Consider going through this activity on your own beforehand to look for potential points of misconception that may occur.

</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Throughout Activity 5.H1.2, students will get the opportunity to engage in individual processing time and work prior to structured discussion with a partner about their findings and conjectures.</td><td>MA.K12.MTR.5.1: Patterns and Structure
This lesson is an extension of what students have learned about congruence in this unit. Students will relate those previously learned concepts to new concepts, while looking for similarities among problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">While this is an honors lesson, student grouping/partnering should be carefully considered based on performance throughout the unit up to this lesson. Some students may be in need of additional scaffolding on support. This lesson provides a great opportunity to clear up misconceptions about congruence and/or specific postulates and theorems.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.H1.1 Warm-Up: Geometry as an Animal

Component Overview: *Students will complete two open-ended questions about geometry as an animal.*

5.H1.1 Warm-Up: Geometry as an Animal

1. If geometry were an animal, it would be a …
   Answers will vary. If geometry were an animal, I think it would be a turtle.

   because …
   Both are majestic and intentional. Turtles do have geometric patterns on their shell. Turtles like Geometry must make every step carefully and intentionally.

2. It would live in …
   Answers will vary. I would live in the ocean.

   because …
   We all know it is out there, however when we do see it, we are surprised and inspired.

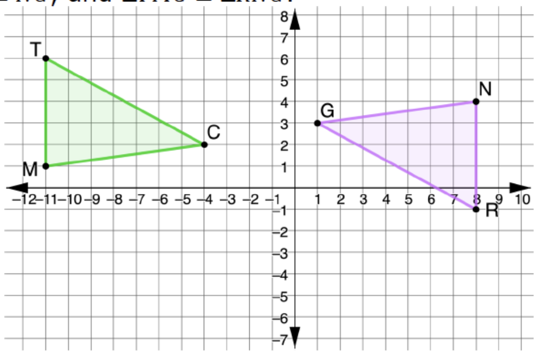
This activity is designed to get students thinking and talking. Consider having students engage in a Think-Pair-Share routine with this activity. Allow students some individual think time and then ask them to work with a partner to share their thoughts and answers with one another.

5.H1.2 Exploration: Justifying Congruence

Component Overview: *Students will explore the concept of congruence further to learn how to justify congruence through three, multi-part, scaffolded questions. For this activity, students will need a partner.*

5.H1.2 Exploration: Justifying Congruence

1. Triangles TCM and RGN are shown, where $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$.



   a. Discuss with your partner what can be concluded from the given information.
   Answers will vary. I can conclude that the triangles are congruent by SAS.

   b. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
   Answers will vary.

Conclusion	My Summary of Their Reason	Initials

When circulating during this activity, listen for academic language. If students are making general observations (such as "one is green and one is blue"), allow those observations, but offer a question that gets them to think about their location or how they may be related to one another.

Add a layer of physical movement to this activity. Consider having students find at least two different students by getting up and moving around the room and finding two unique answers. Consider the use of a timer to maintain instructional momentum and to ensure students don't get lost in the logistics of this activity.

5.H1.2 Exploration: Justifying Congruence (continued)

c. What rigid motions can be applied so that ΔTCM is mapped to ΔRGN?
Answers will vary. Rotation $180°$ about the origin and then translated left 3 and up 5.

d. Discuss with your partner what can be concluded from the rigid motions.
Answers will vary. The shapes are congruent.

e. Ask a classmate for two conclusions they made. Record their conclusions and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

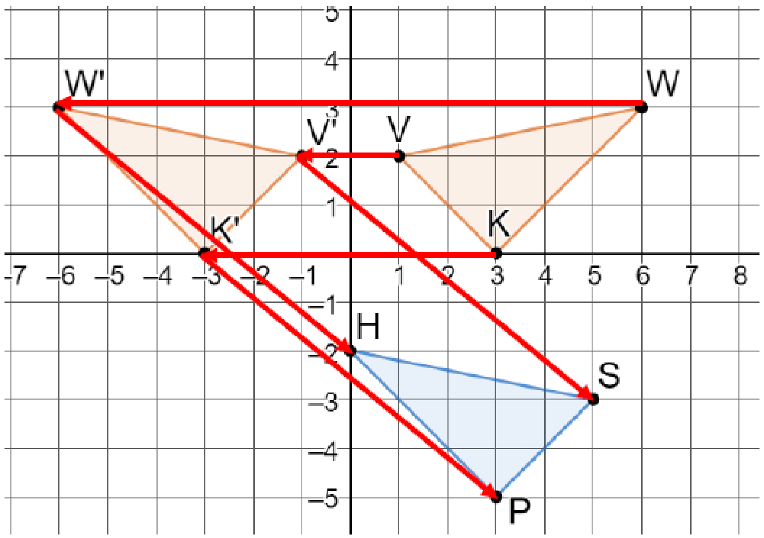
Conclusion	My Summary of Their Reason	Initials

f. Discuss with your partner why showing that the triangles are congruent using the definition of congruence in terms of rigid transformations shows that SAS congruence is sufficient to prove the triangles are congruent. Summarize your discussion.
Answers will vary. Because one triangle was mapped to another using rigid transformations the definition of congruence states that two triangles are congruent. Since $\overline{TM} \cong \overline{RN}$, $\overline{MC} \cong \overline{NG}$, and $\angle TMC \cong \angle RNG$ were given as true then SAS congruence is sufficient to prove the triangles are congruent.

For question 1e, consider adding an additional layer that the two students they use for this table can't be the same two from the earlier part. This will help students engage in diversity of thought and perspective.

5.H1.2 Exploration: Justifying Congruence (continued)

2. Triangles VWK and SHP are shown. Assume $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$.



a. Explain how you could use patty paper to verify $\Delta VWK \cong \Delta SHP$.
I could use patty paper to trace the triangle. Next, I can line the tracing up with the other image to prove that they are congruent.

b. Write a series of transformations that will map ΔVWK onto ΔSHP.
Answers will vary. Reflect ΔVWK across the y-axis, then translate it down 5 right 6.

c. Show each on the coordinate grid.
- Draw $\Delta V'W'K'$, the result of the first transformation
- Map the vertices of ΔVWK to $\Delta V'W'K'$
- Map the vertices of $\Delta V'W'K'$ to ΔSHP

Shown on the graph

For question 2, consider providing students with patty paper so they can use it with question 2a. Instead of just explaining how they could use it to verify, actually ask them to verify it using patty paper.

There are multiple answers that could be correct for question 2b. For students who are moving through the activity quickly, challenge them to come up with an additional series of transformations that could produce the same results.

Consider providing students with a coordinate grid or graph paper to complete question 2c. Otherwise, students may attempt to "sketch" this and lose sight of accuracy.

5.H1.2 Exploration: Justifying Congruence (continued)

d. Complete the statements.

The outcome of mapping the transformations of

ΔVWK to $\Delta V'W'K'$ then from $\Delta V'W'K'$ to ΔSHP shows

that V' coincides with
- ○ H,
- ○ S,
- ○ P,

W' coincides with
- ○ H,
- ○ S,
- ○ P,

and K' coincides with
- ○ H.
- ○ S.
- ○ P.

Because the definition of
- ○ congruence
- ○ similarity

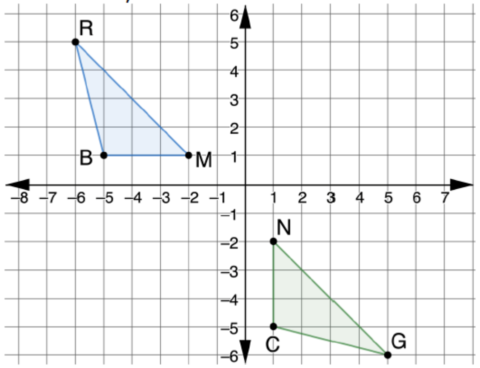
in terms of rigid motions preserves distance and angles, triangle VWK is
- ○ congruent
- ○ similar

to triangle $\underline{SHP}$. Since $\overline{VK} \cong \overline{SP}$, $\overline{WV} \cong \overline{HS}$, and $\overline{KW} \cong \overline{PH}$ was given,
- ○ AAS
- ○ ASA
- ○ SAS
- ○ SSS

is sufficient to show triangles are congruent.

When circulating the room, consider asking students to rephrase what these statements are saying in their own words. Often, students will complete this type of question line by line and not go back to review the concept within the context provided.

3. Triangles RMB and GNC are shown, where $RM \cong GN$, $m\angle BRM \cong m\angle CGN$, and $m\angle BMR \cong m\angle CNG$.



Adriana knows that there is a series of rigid motions that maps ΔRMB onto ΔGNC.

a. Select all true statements after the transformations.
- ☐ Line segment RB coincides with line segment GN.
- ☐ Line segment BM coincides with line segment CN.
- ☐ Line segment CG coincides with line segment BR.
- ☐ Angle NCG coincides with angle RMB.
- ☐ Angle RBM coincides with angle GCN.

b. Explain how the given information and the definition of congruence in terms of rigid motions shows that $\Delta RMB \cong \Delta GNC$.
Answers will vary. It was given that two angles and an included side of one triangle are congruent to two angles and an included side of another triangle. Because the triangles were mapped to another using rigid transformations the definition of congruence states that two triangles are congruent. This shows that ASA is sufficient to show two triangles are congruent.

Consider using a digital tool, such as GeoGebra or Desmos, to allow students to manipulate these triangles in order to explore them and develop their answers to question 3.

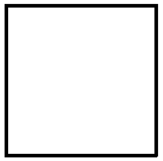
When closing out this component, consider having a whole-group discussion on the answer to question 3b, as it anchors the activity conceptually.

5.H1.3 Wrap-Up: Reflection

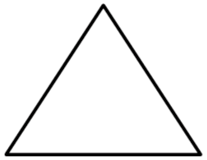
Component Overview: *Students will complete three questions about what they've squared away, three important points, and something circling in their brain.*

5.H1.3 Wrap-Up: Reflection

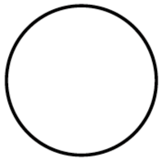
1. What is something that is now "squared away" in your mind about the definition of congruence in terms of rigid transformations?
   Answers will vary.



2. What are three "points" that are important for showing triangle congruence using the definition of congruence in terms of rigid transformations?
   Answers will vary.



3. What is something that is still "circling" your brain about the definition of congruence in terms of rigid transformations?
   Answers will vary.



The information provided from the student here, coupled with what was observed during the exploration, can provide information for differentiating instruction in later lessons.

Consider pairing students to discuss the things still circling their brains to see if peers can answer any of their questions or provide clarification.

5.7 Corresponding Parts of Congruent Triangles Are Congruent

Lesson Introduction

Benchmarks	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Angle, Angle-Angle, and Hypotenuse-Leg. MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.	Learning Targets	• I understand how to use corresponding parts of congruent triangles in a proof. • I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

Benchmark Coherence	Building On	Working Toward
	N/A	N/A

Essential Questions	• How can I prove corresponding parts of congruent triangles are congruent using congruent triangles?
Lesson Narrative	In this lesson, students continue the work they did in the prior lesson on proving that two triangles are congruent. For this lesson, students will include the use of corresponding parts of congruent triangles to prove congruence of other parts. Students will work to complete proofs in various formats to first prove triangle congruence, then to prove the congruence of corresponding parts of congruent triangles.

Component Summary				
	5.7.1 Warm-Up (~5 min.)	Medieval brain teaser *(SE p. 262)*	5.7.4 Guided Practice (5-10 min.)	Practice with completing proofs using CPCTC *(SE p. 266)*
	5.7.2 Collaboration (10 min.)	Collaboration on congruency proofs using overlapping triangles *(SE p. 263)*	5.7.5 Your Turn! (5 min.)	Practice with completing proofs using CPCTC *(SE p. 266)*
	5.7.3 Guided Instruction (10-15 min.)	Formal instruction on completing proofs using CPCTC *(SE p. 264)*	5.7.6 Wrap-Up (~5 min.)	Self-reflection on proving triangles congruent *(SE p. 267)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Corresponding Parts of Congruent Triangles are Congruent (CPCTC) Additional Terms: N/A	• (Optional) Colored Markers • (Optional) Patty Paper	• Students may struggle with the proofs in this lesson. Work through each one on your own prior to delivering instruction to work through potential misconceptions in the proofs.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may not realize all the information that is available in a given problem when it's not explicitly stated. • Students may apply the same marking to multiple sides, causing confusion. • Students may struggle with proving the triangles congruent. • Students may struggle with overlapping triangles.	• Consider having students keep a list of definitions, postulates, and theorems. The list can include common conclusions. For example, if given two lines are parallel, it's common for congruent angles to be the next statement in a proof. • If time is an issue, consider assigning 5.7.5 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Structure 5.7.4 Guided Practice as a Think-Pair-Share. Students can work on their own to try to complete the paragraph proof and then pair up with a partner to determine the correct answers by sharing their thinking and approaches with one another.	**MA.K12.MTR.5.1: Patterns and Structure** Students will be working to complete formal proofs that show two triangles are congruent. They will need to logically order the events in such a way that it solves the problem of proving the two triangles are congruent.
Differentiate Instruction	Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it. For activities that require you to review proofs (including reviewing the answers), consider using auditory explanations for a proof prior to filling in the proof. Use common language to describe how the triangles could be proven congruent. Then, transition that language into formal geometric language when completing the proof. The 5.7.5 Your Turn! provides students an opportunity to demonstrate their understanding of triangle congruence using a proof. Provide students with the opportunity to choose to complete a two-column proof or paragraph proof to demonstrate their understanding. Use their work as a starting point of remediation toward completing flowchart proofs.	

Lesson Notes:

5.7.1 Warm-Up: Brain Teaser

Component Overview: *Students will complete a medieval brain teaser. Students may benefit from drawing a diagram of the man's journey before writing a summary.*

5.7.1 Warm-Up: Brain Teaser

Brain teasers like this have been traced back to medieval times. Here's one with a Florida twist.

1. A man has to take a panther, a raccoon, and some strawberries across St. John's river. His rowboat has enough room for the man, plus either the panther or the raccoon or the strawberries. If he takes the strawberries with him, the panther will eat the raccoon. If he takes the panther, the raccoon will eat the strawberries. Only when the man is present are the raccoon and the strawberries safe from their enemies. All the same, the man successfully carries panther, raccoon, and strawberries across the river. How?
   The man takes the raccoon over to the other side. He goes back and gets the strawberries. Once he delivers the strawberries, he also brings the raccoon back to the first side. He leaves the raccoon and takes the panther back to the other side. He returns one more time to bring the raccoon to the side he is trying to get to.

Watch for the pacing on this activity. Students may get lost in the wording of this brain teaser and want additional time to complete. If students are unable to complete this during the allotted time, consider allowing them time at the end of class to complete.

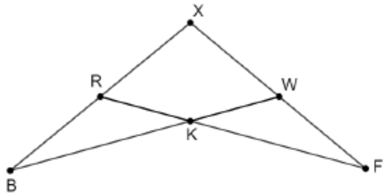
5.7.2 Collaboration: Overlapping Triangles

Component Overview: *Students will work together to complete proofs involving overlapping triangles. For this activity, release students to complete the entire activity with a partner prior to coming back together as a whole class and reviewing the answers together.*

5.7.2 Collaboration: Overlapping Triangles

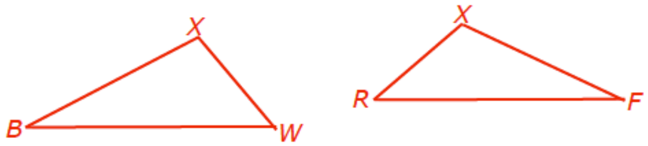
Work with your partner to complete the following.

Polygon $XFKB$ is shown.



1. How many triangles are in polygon $XFKB$?
   4 triangles

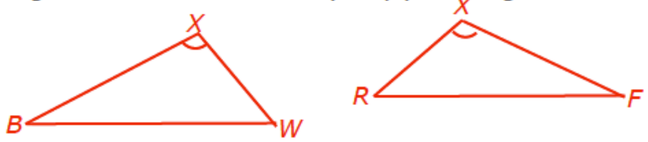
2. Draw ΔXBW and ΔXFR separately.



3. Discuss with your partner if triangles ΔXBW and ΔXFR have any common sides or angles. If so, name the common part(s).
   $\angle X \cong \angle X$

4. Explain why knowing if ΔXBW and ΔXFR have any common sides or angles could be helpful in a proof.
   Answers will vary. Knowing if the triangles have any common sides or angles will identify if there are any congruent parts shown in the figure.

5. In your drawing of ΔXBW and ΔXFR as separate triangles, mark the common part(s) as congruent.

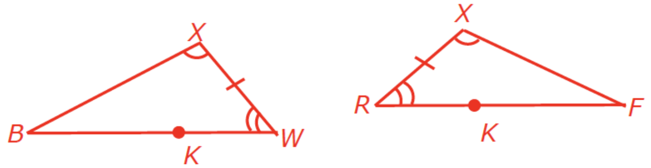


For this activity, allow students to work with their partner to complete all the questions on their own. Consider leveraging this time to work with students who have been struggling with grasping the material in prior lessons.

How can you recognize when the reflexive property of congruence can be used?

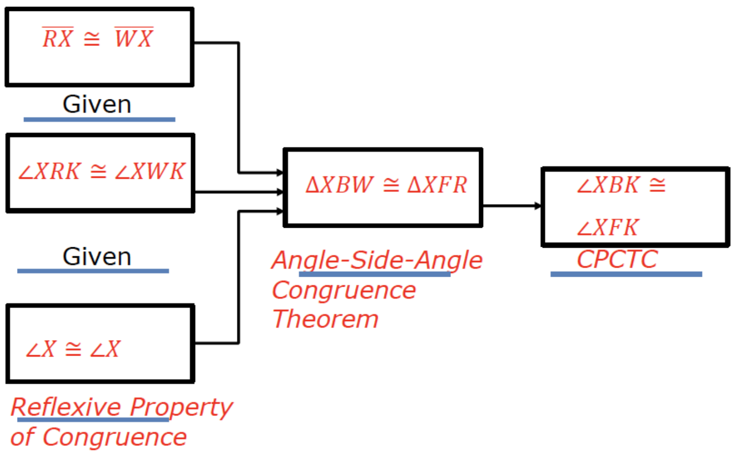
6. What property justifies that something is equal to itself?
   Reflexive Property of Congruence

7. Mark the given information in your diagram.
   Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$



8. What can you conclude about ΔXBW and ΔXFR from the information you marked on your drawing?
   $\Delta XBW \cong \Delta XFR$ by Angle-Side-Angle Congruence Postulate.

9. Complete the flowchart proof for polygon $XFKB$.
   Given: $\overline{RX} \cong \overline{WX}$, $\angle XRK \cong \angle XWK$
   Prove: $\angle XBK \cong \angle XFK$



10. Check your proof with another group and if necessary, make adjustments.

In question 7, students may wonder where point K is in their drawings. Direct students back to the original image to locate K for determining the correct angle.

Consider adding a task in between questions 7 and 8 for this activity. Have student pairs locate an additional student pair to verify their markings to ensure there isn't something they missed.

For a challenge, consider adjusting question 10. Have students trade their proof with another group and be prepared to explain their thinking to an additional group. Each student pair will see two other pairs in this process.

5.7.3 Guided Instruction: Proofs Using CPCTC

Component Overview: *Students will receive guided instruction on completing proofs using CPCTC. For this activity, guide students through all parts of the question and allow students to work with a partner when it's indicated.*

5.7.3 Guided Instruction: Proofs Using CPCTC

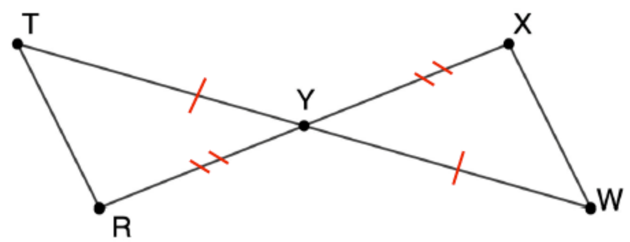
Once two triangles have been proven to be congruent, then CPCTC states that any other corresponding parts of the triangles are also congruent.

Corresponding Parts of Congruent Triangles are Congruent (CPCTC)

If two triangles are congruent, then all corresponding parts of the triangles are congruent.

In the flowchart proof, proving that $\angle XBK \cong \angle XFK$ first required proving that ΔXBW and ΔXFR are congruent.

1. Triangles TYR and XYW are shown.



a. Mark the diagram to show the following given information.
Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
Marked on diagram.

b. What does "Y is the midpoint of $\overline{WT}$" allow you to conclude about the diagram?
"Y is a midpoint of $\overline{WT}$" allows me to conclude that $\overline{TY} \cong \overline{YW}$.

Students have already learned about CPCTC in prior units.

Questions to consider:

- *How can you determine where your markings should go?*
- *What process would you go through to look for additional information that is not listed as "given"?*
- *How can there be additional information that you can mark without it being provided explicitly in the "given"?*

c. What does "Y is the midpoint of $\overline{RX}$" allow you to conclude about the diagram?
"Y is a midpoint of $\overline{RX}$" allows me to conclude that $\overline{RY} \cong \overline{YX}$.

d. Discuss with your partner if there is any other information about the two triangles that can be determined from the diagram. Write a summary of your discussion.
Answers will vary. $\angle XYW \cong \angle TYR$ because of the Vertical Angles Theorem.

e. Work with your partner to complete the paragraph proof.

Given: Y is the midpoint of $\overline{WT}$. Y is the midpoint of $\overline{RX}$.
Prove: $\overline{TR} \cong \overline{XW}$

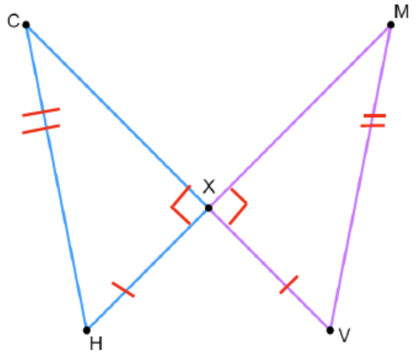
It is given that Y is the midpoint of $\overline{WT}$, so _$\overline{TY} \cong \overline{WY}$_ by __*definition of midpoint*___. Also, it is given that Y is the midpoint of $\overline{RX}$, so _$\overline{RY} \cong \overline{XY}$_ by _*definition of midpoint*_. _$\angle TYR \cong \angle WYX$ by *vertical angles theorem*_. Therefore by _*SAS congruence postulate*_, $\Delta\ TYR \cong \Delta\ WYX$ so by __*CPCTC*_, $\overline{TR} \cong \overline{XW}$.

After releasing students to discuss question 1d with their partners, allow students to share what they find in their discussions. This will give you an opportunity to hear student misconceptions.

For struggling readers or students who may be getting hung up on the paragraph proof, consider adjusting question 1e to include a different proof format such as a flowchart or two-column proof. This will allow you to determine if students conceptually understand the correct components of the proof and may need support on paragraph proofs, or if students may need additional support with the components of the proof themselves.

5.7.3 Guided Instruction: Proofs Using CPCTC (continued)

2. Triangles CXH and XMV are shown.



Given: $\overline{CV} \perp \overline{HM}$, $\overline{HX} \cong \overline{XV}$, $\overline{HC} \cong \overline{MV}$
Prove: $\overline{XC} \cong \overline{XM}$

a. Mark the diagram to show the given information.
Marked on diagram.

b. Which congruence theorem or postulate do you think should be used?
HL Congruence

c. Complete the two-column proof.

Statement	Reason
1. $\overline{CV} \perp \overline{HM}$	1. Given
2. $\overline{HX} \cong \overline{XV}$	*2. Given*
3. $\overline{HC} \cong \overline{MV}$	*2. Given*
4. $\angle CXH$ and $\angle MXV$ are right angles	*4. Definition of Perpendicular Lines*
5. ΔCXH and ΔMXV are right triangles	*5. Definition of right triangle*
6. $\Delta CXH \cong \Delta MXV$	*6. Hypotenuse Leg Theorem*
7. $\overline{XC} \cong \overline{XM}$	*7. CPCTC*

Prior to teaching through question 2c, consider having students do a "brain storm" of everything they know and observe about these two triangles. This can help students get all their observations out and then be able to organize them into a formal two-column proof.

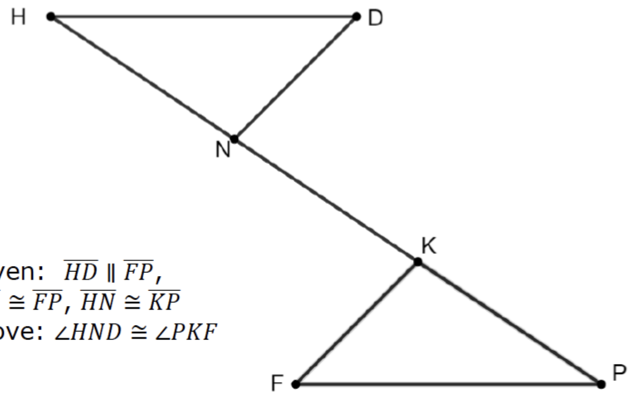
Why is SSA not something that could be used, but HL can? Think back to previous lessons.

5.7.4 Guided Practice: Proofs Using CPCTC

Component Overview: *Students will engage in a structured practice for completing triangle congruency proofs using CPCTC. For this activity, provide students with a structured opening of reviewing the image and doing the markings. Allow students to try to complete the rest on their own prior to reviewing the correct answers.*

5.7.4 Guided Practice: Proofs Using CPCTC

1. A diagram of triangles NDH and PKF with $\overline{NK}$ is shown.



Given: $\overline{HD} \parallel \overline{FP}$,
$\overline{HD} \cong \overline{FP}$, $\overline{HN} \cong \overline{KP}$
Prove: $\angle HND \cong \angle PKF$

Complete the paragraph proof.

It is given that $\overline{HD} \parallel \overline{FP}$, so by the Alternate Interior Angles Postulate, $\angle DHN \cong \angle FPK$. Since it is given that $\overline{HD} \cong \overline{FP}$ and $\overline{HN} \cong \overline{KP}$, $\Delta DHN \cong \Delta FPK$ by SAS.

Therefore, $\angle HND \cong \angle PKF$ by CPCTC.

For this activity, consider beginning by doing a whole-class review of the given information. Allow students to determine what information is explicitly given and what additional markings they can make based on other information. Once this is completed, allow students to work on their own to complete the fill-in-the-blank components.
Think-Pair-Share
Students can work on their own to try and complete the paragraph proof and then pair up with a partner to determine the correct answers by sharing their thinking and approaches with one another.

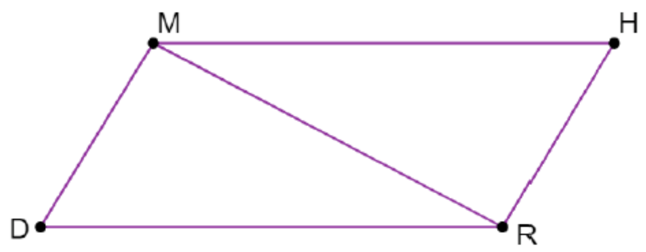
While students are working and during the review of the answer, pay close attention to students who are not grasping the content. Collect that data during this activity to be able to provide differentiated support during the next activity.

5.7.5 Your Turn!

Component Overview: *Students will demonstrate their knowledge of completing a proof using CPCTC. If students have been keeping an organized list of definitions, postulates, and theorems, allow them to reference that resource during this activity.*

5.7.5 Your Turn!

1. Quadrilateral $MHRD$ is shown.



Given: $\overline{DM} \parallel \overline{RH}$, $\angle RDM \cong \angle RHM$
Prove: $\overline{DM} \cong \overline{RH}$

Complete the flowchart proof.

$\overline{DM} \parallel \overline{RH}$ — Given

$\angle DMR \cong \angle HRM$ — Alternate Interior Angles Postulate

$\angle RDM \cong \angle RHM$ — Given

$\overline{MR} \cong \overline{MR}$ — Reflexive Property of Congruence

$\Delta DMR \cong \Delta HRM$ — AAS Congruence Postulate

$\overline{DM} \cong \overline{RH}$ — CPCTC

For differentiation, provide students with the opportunity to choose to complete a two-column proof or paragraph proof to demonstrate their understanding. Use their work as a starting point of remediation toward completing flowchart proofs.

5.7.6 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on student understanding of the content of this lesson. This activity should be done individually to provide data on individual student perception.*

5.7.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I understand how to use corresponding parts of congruent triangles in a proof.
   Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

2. I can prove parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.

Consider having students mark their reflections on each of the different components in the lesson.

5.8 The Proof Is in the Bag!

Lesson Introduction

<table>
<tr><td rowspan="1">Benchmarks</td><td>MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.
MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.
MA.912.LT.4.10 Judge the validity of arguments and given counterexamples to disprove statements.</td><td>Learning Targets</td><td>I can prove that two triangles are congruent using SSS, SAS, ASA, AAS, or HL congruence.
I can prove that parts of triangles are congruent using triangle congruence and corresponding parts of congruent triangles.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td>Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td>N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="3">How can I prove that two triangles are congruent?
How can I prove that parts of triangles are congruent?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="3">In this lesson, students will apply what they have learned about triangle congruency to complete formal proofs. Students will engage in a culminating experience with triangle congruency.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>5.8.1 Warm-Up (~5 min.)</td><td colspan="2">Two-column proof about triangle congruency (SE p. 268)</td></tr>
<tr><td>5.8.2 Activity (35 min.)</td><td colspan="2">Card sort activity to complete given proofs using given statement and reason cards (SE p. 268)</td></tr>
<tr><td>5.8.3 Wrap-Up (~5 min.)</td><td colspan="2">Students summarize their learning (SE p. 271)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td>Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms</u>: N/A

<u>Additional Terms</u>: N/A</td><td>Card sort</td><td>Work through each of these proofs prior to facilitating the activity. Students may struggle with certain ones and it will be important that you know the proofs ahead of time in order to provide guidance.
Visit our online teacher area to download a copy of the card sort to prepare for component 5.8.2.</td></tr>
</table>

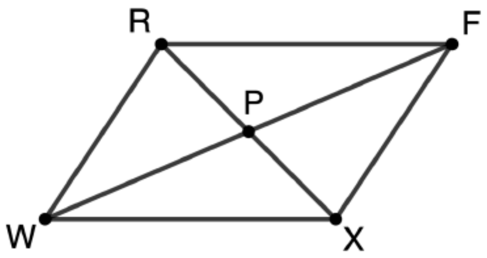
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with ordering the statements.
Student may apply the same marking to multiple sides, causing confusion.
Students may misinterpret or mislabel corresponding parts of triangles.
Students may struggle with matching statements with the correct reasons.</td><td>This activity is a culminating activity that focuses on proofs. With multiple proofs occurring with multiple cards, it would be beneficial to use colored paper for each proof to keep things sorted as you facilitate the activity.
Students may forget this could be different for some statements. It may help to ask questions about whether the statement relies on another statement or if it is independent.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Stronger and Clearer Each Time
5.8.2 Activity has students working on multiple different proofs as a group. Consider leveraging the “stronger and clearer each time” protocol to provide a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output.</td><td>MA.K12.MTR.5.1: Patterns and Structure
For 5.8.2 Activity, students will be working together to determine the correct steps and reasons to complete proofs that show that two given triangles are congruent, or parts of triangles are congruent. Students will need to create plans to logically order the steps in the proof.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider placing students in groups based on their performance so far in this unit. Students will be applying their knowledge from prior lessons through the process of completing a proof. Students who struggled with triangle congruency will likely struggle proving it.
For activities that require you to review proofs (including reviewing the answers), consider using auditory explanations for a proof prior to filling it in. Use common language to describe how the triangles could be proven to be congruent. Then, transition that language into formal geometric language when completing the proof.
Consider altering the assignment to eliminate Proof C and Proof F for students who need more time to complete the card sort.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.8.1 Warm-Up: Bell Work

Component Overview: *Students will complete a two-column proof about triangle congruency. For this activity, students should work individually in order for the instructor to get an idea of where students are entering this content.*

5.8.1 Warm-Up: Bell Work

Polygon $RFXW$ where diagonals $\overline{RX}$ and $\overline{FW}$ bisect each other at point P is shown.



Given: $\overline{RX}$ and $\overline{FW}$ bisect each other at point P.

Prove: $\Delta RPF \cong \Delta XPW$

1. Complete the two-column proof.

Statement	Reason
1. $\overline{RX}$ and $\overline{FW}$ bisect each other at point P	1. Given
2. $RP = PX$ and $WP = PF$	2. Definition of segment bisector
3. $\overline{RP} \cong$ _$\overline{XP}$_ and _$\overline{FP}$_ $\cong \overline{WP}$	3. Definition of congruence
4. $\angle RPF \cong \angle XPW$	4. *Vertical angles are congruent*
5. $\Delta RPF \cong \Delta XPW$	5. *SAS Congruence*

The data from this Warm-Up can provide data for which students may need additional and immediate support as the next activity launches to ensure they get off on the right track.

5.8.2 Activity: It's in the Bag!

Component Overview: *For this activity, students will work in small groups to complete each of the proofs, using the cards needed for this activity. If students have previously created a list of definitions, postulates, and theorems, allow them to use this during the activity.*

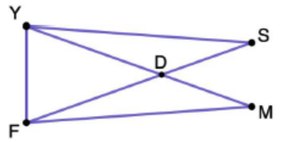
5.8.2 Activity: It's in the Bag!

With your group, complete each proof using the statement and reason cards provided.

Proof A

Given: $\angle YSD \cong \angle FMD$
ΔYDF is isosceles with base $\overline{YF}$

Prove: $\overline{FM} \cong \overline{YS}$

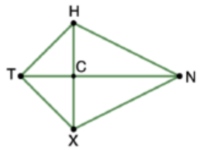


Statement	Reason
$\angle YSD \cong \angle FMD$	Given
ΔYDF is isosceles with base $\overline{YF}$	Given
$\overline{YD} \cong \overline{FD}$	Definition of isosceles triangle
$\angle SDY \cong \angle MDF$	Vertical angles are congruent
$\Delta DMF \cong \Delta DSY$	AAS congruence theorem
$\overline{FM} \cong \overline{YS}$	CPCTC

Proof B

Given: $\overline{TN}$ bisects HTX
$\overline{HT} \cong \overline{XT}$

Prove: $\overline{TN}$ bisects $\angle HNX$



Statement	Reason
$\overline{HT} \cong \overline{XT}$	Given
$\overline{TN}$ bisects $\angle HTX$	Given
$m\angle HTC = m\angle XTC$	Definition of angle bisector
$\angle HTC \cong \angle XTC$	Definition of congruence
$\overline{TN} \cong \overline{TN}$	Reflexive property of congruence
$\Delta TXN \cong \Delta THN$	SAS congruence postulate
$\angle HNT \cong \angle XNT$	CPCTC
$m\angle HNT = m\angle XNT$	Definition of congruence
$\overline{TN}$ bisects $\angle HNX$	Definition of angle bisector

There is a blackline master for this collaborative activity to happen at stations, through a gallery walk, or in traditional small groups.

Prior to the lesson, cut out the statement/reasons from the black-line master and place in plastic bags for students to use when completing proofs. Consider using different-colored paper or other visual indicators for each proof to maintain control over all the correct pieces.

This activity could be done digitally through a platform that provides card sort activities such as Desmos, Padlet, Kahoot etc.

Stronger and Clearer Each Time
Consider leveraging the stronger and clearer each time protocol to provide a structured and interactive opportunity for students to revise and refine both their ideas and their verbal and written output.

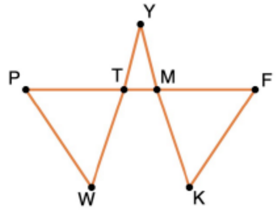
Consider developing a checklist for your use as you observe students working on the proofs.

5.8.2 Activity: It's in the Bag! (continued)

Proof C

Given: $\angle WPT \cong \angle KFM$
$\overline{PW} \cong \overline{FK}$
$\angle YMT \cong \angle YTM$

Prove: $\Delta PWT \cong \Delta FKM$

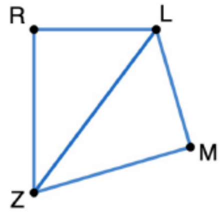


Statement	Reason
$\angle WPT \cong \angle KFM$	Given
$\overline{PW} \cong \overline{FK}$	Given
$\angle YMT \cong \angle YTM$	Given
$\angle YMT \cong \angle FMK$	Vertical angles are congruent
$\angle YTM \cong \angle PTW$	Vertical angles are congruent
$\angle FMK \cong \angle PTW$	Transitive property of congruence
$\Delta PWT \cong \Delta FKM$	AAS congruence theorem

Proof D

Given: $\overline{LM} \cong \overline{LR}$
$\overline{LR} \perp \overline{RZ}$
$\overline{LM} \perp \overline{MZ}$

Prove: $\angle MZL \cong \angle RZL$



Statement	Reason
$\overline{LR} \perp \overline{RZ}$, $\overline{LM} \perp \overline{MZ}$	Given
$\angle LMZ$ and $\angle LRZ$ are right angles	Definition of perpendicular lines
ΔLMZ and ΔLRZ are right triangles	Definition of right triangle
$\overline{LM} \cong \overline{LR}$	Given
$\overline{LZ} \cong \overline{LZ}$	Reflexive property of congruence
$\Delta LMZ \cong \Delta LRZ$	HL theorem
$\angle MZL \cong \angle RZL$	CPCTC

What are some things that you notice that are different about the type of image in Proof C than other images in other proofs you have worked with?

Students often confuse transitive property of congruence and substitution property of equality. Substitution property can apply to statements of equality while transitive property can apply to statements of congruence.

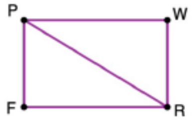
When sharing the correct answers for this activity, consider having groups present their findings, requiring that they also include their explanation and rationale for their answer choices.

5.8.2 Activity: It's in the Bag! (continued)

Proof E

Given: $\overline{PF} \cong \overline{WR}$
$\overline{PW} \cong \overline{FR}$

Prove: $\overline{PF} \parallel \overline{WR}$

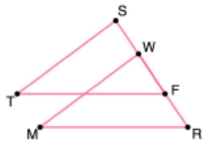


Statement	Reason
$\overline{PF} \cong \overline{WR}$	*Given*
$\overline{PW} \cong \overline{FR}$	*Given*
$\overline{PR} \cong \overline{PR}$	*Reflexive property of congruence*
$\Delta PRF \cong \Delta RPW$	*SSS congruence postulate*
$\angle FPR \cong \angle WRP$	*CPCTC*
$\overline{PF} \parallel \overline{WR}$	*If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.*

Proof F

Given: $\overline{RF} \cong \overline{SW}$
$\angle TSF \cong \angle MWR$
$\overline{FT} \parallel \overline{MR}$

Prove: $\Delta TSF \cong \Delta MWR$



Statement	Reason
$\overline{RF} \cong \overline{SW}$	*Given*
$RF = SW$	*Definition of congruence*
$FW = FW$	*Reflexive property of equality*
$RF + FW = SW + FW$	*Addition property of equality*
$RF + FW = RW$ and $SW + WF = SF$	*Segment addition postulate*
$RW = SF$	*Substitution property*
$\overline{RW} \cong \overline{SF}$	*Definition of congruence*
$\overline{FT} \parallel \overline{MR}$	*Given*
$\angle TFS \cong \angle MRW$	*If two parallel lines are intersected by a transversal, then corresponding angles are congruent.*
$\angle TSF \cong \angle MWR$	*Given*
$\Delta TSF \cong \Delta MWR$	*ASA congruence theorem*

Consider using some of the time during this activity to pull small groups of students aside to provide additional support based on their performance with this content.

For an additional challenge, consider having students write the proof in a different format, such as a flowchart proof or a paragraph proof.

5.8.3 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

5.8.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about writing proofs.
   Answers will vary.

Consider using the learning targets as sentence frames for students to reflect on using content from the lesson. Alternatively, provide sentence frames specific to a certain concept that you would like additional data to inform differentiation in future lessons.

5.9 Congruence in Two-Dimensional Figures – Part 1

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.</td><td>Learning Target</td><td>I can solve mathematical problems involving congruence.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How can I use my knowledge of triangle congruence to solve mathematical problems?
How can corresponding parts of congruent polygons be used to determine measures?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students will apply their knowledge of triangle congruence and corresponding parts of congruent triangles from the prior lessons in this unit to solve mathematical problems. Students will work together to learn how the use of CPCTC can help to determine measures of a given triangle and extend that into understanding corresponding parts of congruent polygons are congruent. They will then use this knowledge to solve mathematical problems.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>5.9.1 Warm-Up
(~5 min.)</td><td>Complete a two-column proof on triangle congruence (SE p. 272)</td><td>5.9.3 Guided Instruction
(10-15 min.)</td><td>Apply corresponding parts of congruent polygons to find unknown angles and lengths (SE p. 273)</td></tr>
<tr><td>5.9.2 Collaboration
(10 min.)</td><td>Collaborate on using CPCTC to determine unknown measures (SE p. 273)</td><td>5.9.4 Guided Practice
(10-15 min.)</td><td>Use corresponding parts to solve mathematical problems (SE p. 275)</td></tr>
<tr><td>5.9.5 Wrap-Up
(~5 min.)</td><td colspan="3">Error analysis activity on congruent parts to solve a mathematical problem (SE p. 276)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms</u>:
Corresponding Parts of Congruent Polygons are Congruent
<u>Additional Terms</u>: N/A</td><td colspan="2">(Optional) Markers
(Optional) Patty paper</td><td>Some of these problems have areas where minor missteps can throw off the entire process. Work through each of these problems on your own, prior to teaching, to look for these errors that students may make.</td></tr>
</table>

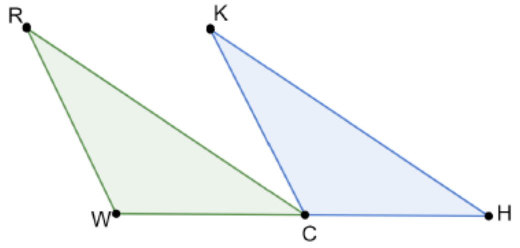
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with decomposing the figures.
Students may not understand which parts in congruent polygons are corresponding parts.
Students may not recall how to use the Pythagorean Theorem.</td><td>This lesson is the first of a two-part series. Each problem will require students to use given information to make markings as well as additional information, such as the Pythagorean Theorem.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
5.9.2 Collaboration is a collaborative activity for students to work together on using CPCTC to determine measures. In this activity, students will have the opportunity to think about the given triangle and then pair up with a partner to discuss their work and come to a collaborative solution.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
During the Wrap-Up activity, students will complete an error analysis activity on congruent parts to solve a mathematical problem.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider partner pairings by performance on the lessons prior to the proof sections. Placing students together who struggle with the same concept can provide you with an opportunity to work with smaller groups on similar issues.
Consider using manipulatives or other tools to deconstruct certain polygons to help students visualize the shapes that comprise them.
5.9.4 Guided Practice has two problems. Consider having struggling students only work on question 1 that has been edited with additional scaffolding included. Samples of scaffolding questions are:
What type of triangle is ΔYHD?
What formula can be used to find a missing side of a right triangle?
$\angle HDT$ has the same measure as which angle?
What is the sum of the measures of the three angles of a triangle?</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.9.1 Warm-Up: Bell Work

Component Overview: *Students will fill in the missing statements and reasons in a two-column proof on triangle congruence. For this activity, allow students to work independently to determine what they recall from the prior lesson.*

5.9.1 Warm-Up: Bell Work

Triangles RCW and KHC are shown.



Given: $RW \parallel KC$, C is the midpoint of $\overline{WH}$, $\overline{WR} \cong \overline{CK}$
Prove: $\angle WRC \cong \angle CKH$

1. Complete the missing statements and reasons in the two-column proof.

Statement	Reason
1. C is the midpoint of $\overline{WH}$	1. Given
2. $\overline{WC} \cong \overline{CH}$	2. Definition of midpoint
3. $RW \parallel KC$	3. Given
4. $\angle RWC \cong \angle KCH$	4. *Corresponding Angles Theorem*
5. $\overline{WR} \cong \overline{CK}$	5. Given
6. $\Delta RWC \cong \Delta KCH$	6. *SAS Congruence Postulate*
7. $\angle WRC \cong \angle CKH$	7. *CPCTC*

For this activity, allow students to try this problem completely on their own to see what they have retained from the previous lesson. Use the information from the formative assessment to determine possible misconceptions your students have.

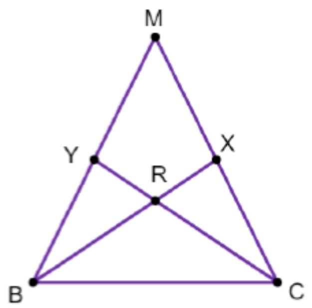
Consider adjusting this Warm-Up based on student performance in the prior lesson. If students are struggling with the content, consider providing a word bank for them to use to match to the reason only.
If students are doing well, consider adding the challenge of having them complete the whole proof without prompts.

5.9.2 Collaboration: Using CPCTC to Determine Measures

Component Overview: *Students will work together to use CPCTC to determine unknown measures. For this activity, students should work with a partner or small group to move through discovery of the content.*

5.9.2 Collaboration: Using CPCTC to Determine Measures

Triangle MBC is shown where $MX = 3.6$, $MB = 7.2$, $RX = 2$, $BC = 6.5$, $RB = 3.9$, $m\angle YBR = 29°$, $m\angle CXR = 83°$, and $m\angle MCB = 63°$.



Work with your partner to complete the following.

1. How many different triangles are in ΔMBC?
   7 Triangles

2. If $\Delta BRY \cong \Delta CRX$, then determine the measure of each.
   $RC =$ ___3.9___ $m\angle RYB =$ ___83°___

3. If $\Delta BXM \cong \Delta CYM$, then determine the measure of each.
   $CY =$ ___5.9___ $CM =$ ___7.2___ $m\angle MCY =$ ___29°___

Visual learners may benefit from using colored markers or highlighters to help visualize different angles and side lengths when doing their markings within the triangle.

When reviewing the answers with students, allow them to share out their thinking to each question. For question 1, consider having each student share one of the 8 triangles that they found.

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

Component Overview: *Students will receive guided instruction on corresponding parts of congruent polygons.*

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

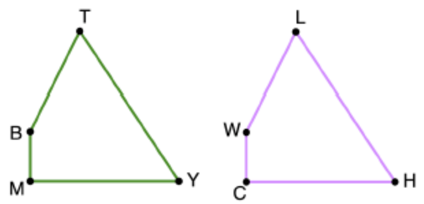
Any polygon that can be mapped to another polygon using rigid transformations shows that all of the parts of the polygons are congruent. This understanding can then be applied to find measurements of the two congruent polygons.

Corresponding Parts of Congruent Polygons are Congruent

If two polygons are congruent, then all corresponding parts of the polygons are congruent.

1. Congruent quadrilaterals $TYMB$ and $LHCW$ are shown.

   Some measurements of the angles in quadrilaterals $TYMB$ and $LHCW$ are given.



$m\angle YMB = 90°$,
$m\angle WLH = 60°$, $m\angle MBT + m\angle CHL = 210°$, $m\angle BTY = (2x - 4)°$,
$m\angle MYT = (x + 25)°$

   a. Determine each.

   $m\angle BTY$

   60°

   $m\angle CWL$

   153°

Students should be familiar with the corresponding parts concepts from prior lessons with triangles. Emphasize this concept when introducing CPCTC and the connection to the definition of congruence, which states that if figures are congruent, then they have equal measures. Reminding students of rigid motions on the coordinate plane may also help students recall that congruent figures would have corresponding parts with equal measures.

What markings did you make in the quadrilaterals with the given information?

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons (continued)

Some measurements of the sides of quadrilaterals $TYMB$ and $LHCW$ are given.

$HC = 5k$, $LW = 4k - 0.5$, $YT = 6k$, and $BM = k + 1$.

b. Determine the value of k if the perimeter of quadrilateral $TYMB$ is 24.5.

$$k + 1 + 4k - 0.5 + 6k + 5k = 24.5$$

$$k = 1.5$$

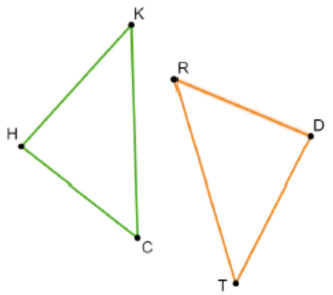
c. What is the length of BT?

$$BT = 4k - 0.5$$

$$BT = 5.5$$

Students should be familiar with perimeter, but still consider asking students what that word means. This can benefit ELL students or students who are unfamiliar with the concept. Monitor students' work on solving the equation to determine if remediation is needed for the algebra or if there is a misunderstanding of the geometry in the question.

2. The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$. Some measurements of the sides are given. $KH = 5y - 2$, $DR = 6y - 5$, and $TD = 4y + 3$



Arthur's Work
$5y - 2 = 6y - 5$ $-2 = y - 5$ $3 = y$
$KH = 5(3) - 2$ $KH = 13$

Arthur used the information to determine the length of KH. His work is shown.

a. Find Arthur's error.

$\overline{KH} \cong \overline{TD}$ because corresponding parts of congruent polygons are congruent. Arthur used the wrong side when setting up his equation. He used side length RD instead of TD.

For question 2, consider allowing students to try this one on their own first after introducing the context. If needed, assist students with marking the triangles using the given information and then releasing them. When reviewing, have students share their rationale and leverage an agree/disagree protocol.

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons (continued)

b. Help Arthur understand and learn from his mistake. Write a brief note to Arthur explaining what his mistake was and how he can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

Provide sentence stems, such as:

Arthur made a mistake when he ______ because______.

Arthur should have ________ instead of _________.

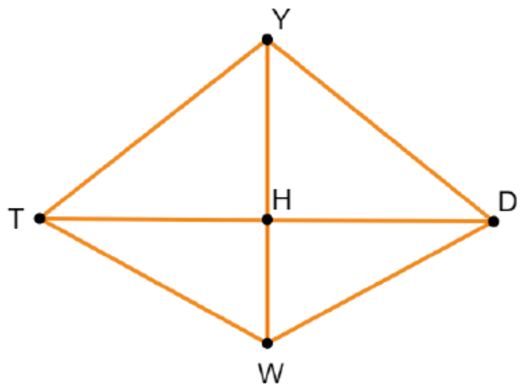
The next time Arthur sees ______, Arthur should ______.

5.9.4 Guided Practice: Corresponding Parts

Component Overview: *Students will engage in structured practice on corresponding parts to solve mathematical problems.*

5.9.4 Guided Practice: Corresponding Parts

1. Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



   a. How many different triangles are in Quadrilateral $YDWT$?
   8 *Triangles*

   b. If $\Delta DHY \cong \Delta THY$, $YD = \sqrt{130}$, $YH = 7$, and $m\angle HTY = 38°$, determine the following measures.

   $HD =$ __9__ $m\angle DYH =$ $\underline{52^{\circ}}$ $TH =$ ___9___

   c. If $\Delta DHW \cong \Delta THW$, $WH = 5$, and $m\angle HWD = 61°$, use these and all other known measurements to determine the following measures.

   $m\angle TWH =$ $\underline{61^{\circ}}$ $m\angle HTW =$ _$\underline{29^{\circ}}$_ $TW =$ _$\underline{\sqrt{106}}$_

Consider having struggling students only work on question 1 that has been edited to include additional scaffolding. Samples of scaffolding questions are,

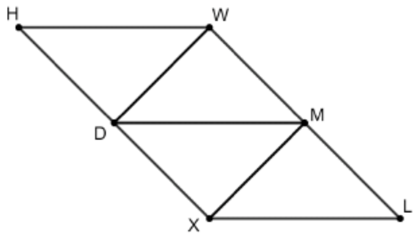
- *What type of triangle is ΔYHD?*
- *What formula can be used to find a missing side of a right triangle?*
- *$\angle HDT$ has the same measure as which angle?*
- *What is the sum of the measures of the three angles of a triangle?*

In question 1c, students may think that there is not enough information to find TW. Remind them that they can use the Pythagorean Theorem to find the length of HD, which can then be used to find the length of DW, which will be congruent to TW based on all the information given.

5.9.4 Guided Practice: Corresponding Parts (continued)

Component Overview: *Students will complete an error analysis activity on congruent parts to solve a mathematical problem. This activity is designed to be done independently to get student data on their level of comprehension of the content of this lesson.*

2. Quadrilateral $HWLX$ is shown.



$HWMX \cong LXDW$ and $\Delta XMD \cong \Delta WDM$. The perimeter of $HWMX$ is 40.5.

a. Use the given measurements to find the value of x.
$HW = 2x - 2$, $XD = \frac{3}{2}x - 1$, $HX = 2x + 5$, $WD = 2x - 2$

$$2x - 2 + \frac{3}{2}x - 1 + 2x + 5 + 2x - 2 = 40.5$$

$$x = 5.4$$

b. Find the length of MX.

$$MX = WD = 2x - 2$$

$$MX = 8.8$$

Students may struggle to visualize the image provided in question 2. Encourage students to first draw the two quadrilaterals separately and/or to mark the congruent pieces from the congruency statements.

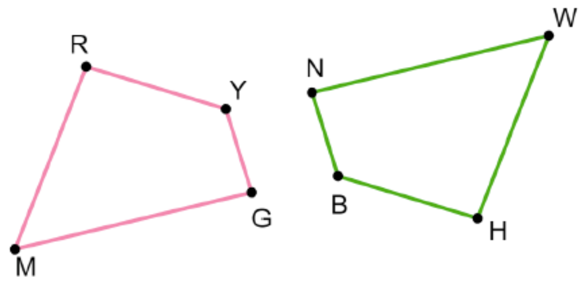
For an additional challenge, consider having students find the length of additional sides, including the full perimeter of the quadrilateral.

5.9.5 Wrap-Up: Error Analysis

Component Overview: *Students will complete an error analysis on congruent parts to solve a mathematical problem. This is designed to be done independently to get student data on their level of comprehension of the content of this lesson.*

5.9.5 Wrap-Up: Error Analysis

1. Quadrilaterals $RYGM$ and $HBNW$ are shown.



$RYGM \cong HBNW$ and the following information is also true.

$m\angle RYG = (10x - 6)°$, $MG = 2k + 3$, $NW = 4k - 29$,
$m\angle MGY = (7x - 5)°$, $m\angle BNW = 86°$, $HB = k + 9$

Hector used the given information to write the two equations shown.

$$10x - 6 = 86 \qquad\qquad 4k - 29 = 2k + 3$$

One of Hector's equations is NOT correct. Determine which one, and correct his error.
The first equation is incorrect. The equation should be $7x - 5 = 86$

This question provides an error analysis on the content from this lesson. The data from this activity can provide information on whether or not a student understands, and to what degree, the content from this lesson.

5.10 Congruence in Two-Dimensional Figures – Part 2

<table>
<tr><th colspan="9">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="3">MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.</td><td colspan="2">Learning Target</td><td colspan="3">• I can solve real-world problems involving congruence.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="4">Building On</td><td colspan="4">Working Toward</td></tr>
<tr><td colspan="4">N/A</td><td colspan="4">N/A</td></tr>
<tr><td>Essential Question</td><td colspan="8">• How can I use my knowledge of congruence of two figures to solve real-world problems?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="8">In this lesson, students will extend their learning from the prior lesson (Part 1) to apply it to real-world scenarios. Students will learn about corresponding parts in the real world and then use that knowledge to solve real-world problems that involve congruency.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>5.10.1 Warm-Up (~5 min.)</td><td colspan="3">Menu pricing Notice and Wonder activity (SE p. 277)</td><td colspan="2">5.10.3 Collaboration (10-15 min.)</td><td colspan="2">Collaboration with solving real-world problems using corresponding parts (SE p. 278)</td></tr>
<tr><td>5.10.2 Guided Instruction (10-15 min.)</td><td colspan="3">Identifying and using corresponding parts in the real world (SE p. 277)</td><td colspan="2">5.10.4 Your Turn (5-10 min.)</td><td colspan="2">Solving a real-world problem using corresponding parts (SE p. 279)</td></tr>
<tr><td>5.10.5 Wrap-Up (~5 min.)</td><td colspan="7">Self-reflection on the different concepts of this unit (SE p. 280)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td colspan="4">Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms</u>: N/A

<u>Additional Terms</u>: N/A</td><td colspan="2">• (Optional) Markers</td><td colspan="4">• Some of these problems can be more challenging to decipher than others. Work through each problem on your own prior to delivering this lesson to check for potential misconceptions.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may struggle with reading or relating to the context.
Students may not understand which parts in congruent polygons are corresponding parts.
Students may not translate the question correctly using the provided information.</td><td>This lesson is the second of a two-part series. Each problem will require students to use word problem and reading comprehension skills in addition to the necessary mathematics. Consider a review of a word problem strategy prior to launching the lesson.
Students learned about interior angles sums in 8^{th} grade Pre-Algebra. However, there may be students who do not remember. Consider a quick review (no more than 2 minutes) of decomposing polygons into triangles to find the sum of the angles.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
5.10.1 Warm-Up asks students to determine things they notice and things they wonder about items on a menu, using the various values of different components.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Throughout this lesson, students will be provided with complex problems about congruency through corresponding parts in real-world scenarios. Students will learn and practice decomposing these problems into manageable parts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For a different entry point in the Warm-Up, provide specific scaffolding questions that will guide the students to what can be noticed when comparing the prices.
Prior to launching the 5.10.2 Guided Instruction, consider reviewing your chosen word problem strategy with students, modeling its use in the problems used during instruction.
The Wrap-Up provides students with a self-reflection on the concepts of the entire unit. Use their responses, coupled with their overall performance, to determine which students may need additional support.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.10.1 Warm-Up: Notice and Wonder

Component Overview: *Students will complete a notice and wonder activity about menu prices.*

5.10.1 Warm-Up: Notice and Wonder

A picture of a menu is shown.

Burgers	Chips	Fries
Deluxe Burger	$6.90	$7.70
Deluxe Cheese	$7.70	$8.45
Bacon Burger	$7.95	$8.65
Bacon Cheese	$8.25	$8.95
Double Burger	$8.35	$9.30
Double Cheese	$9.25	$9.95
Double Bacon Cheese	$9.50	$10.25
Cheese Garden Burger	$8.25	$8.85
Mushroom Swiss	$8.25	$8.95
Jalapeno Pepper Jack	$8.25	$8.95
BBQ Western Bacon	$9.20	$9.85
Pastrami Burger	$9.20	$9.85
Power Pac	$10.45	$11.05
Cheese Power Pac	$11.25	$11.70

1. What do you notice about prices of the burgers with cheese?
   Answers will vary. The burgers with cheese have higher prices, but not by the same amount.
2. What do you notice about ordering chips versus fries?
   Answers will vary. Chips are less than fries, but also by different amounts.
3. There is a price difference between adding chips or fries with a double burger and adding chips or fries with cheese garden burger. Do you think this means that when you order a double burger you get more fries?
   Answers will vary. I would not expect that this has anything to do with the amount of fries, however, maybe more people add fries to the double burger, so the company can make more money by charging more compared to the Cheese garden burger.
4. When you look at a menu, do you consider the price differences or just order what you would like to eat?
   Answers will vary. I use to look at all prices, find the lowest few items, and select from them. Now I have started picking a few things I want to eat, then compare the cost.

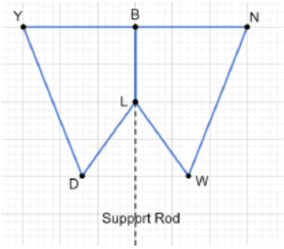
For a different entry point, provide questions to students that highlight specific things to notice. For example:

- *What is the price difference between a Deluxe Burger and a Deluxe Cheese?*
- *What is the price difference between a Bacon Burger and a Bacon Cheese?*
- *What is the price difference between ordering a Deluxe Burger with Fries and ordering it with Chips?*
- *What is the price difference between ordering a Bacon Burger with Fries and ordering it with Chips?*

5.10.2 Guided Instruction: Corresponding Parts in the Real World

Component Overview: *Students will receive guided instruction on how to identify and use corresponding parts in the real world.*

5.10.2 Guided Instruction: Corresponding Parts in the Real World



Oscar and Aarna worked together to create a prototype obstacle for a drone obstacle course. They used grid paper to create their design as shown.

To be sure the design met regulations for the competition, they used the equations to input the design into a computer program. They discovered that some of their work disappeared, so they used the given equations knowing that $YBLD \cong NBLW$.

The perimeter of $YBLD$ is 293 centimeters (cm). $\angle YBL$ is a right angle.

$YB = 75$ cm
$WL = (3x + 10)$ cm
$m\angle LWN = (4k + 9)°$
$m\angle DLB = (12k + 1)°$
$WN = (6x + 5)$ cm
$LB = 50$ cm
$m\angle BYD = (5k + 8)°$

1. What is the value of x?
   $75 + 6x + 5 + 3x + 10 + 50 = 293$
   $x = 17$
2. What is the length of WL? Include units.
   $WL = 3x + 10 = 61$ *cm*
3. What is the length of YD? Include units.
   $YD = WN = 6x + 5 = 107$ *cm*
4. What is the value of k?
   $5k + 8 + 90 + 12k + 1 + 4k + 9 = 360$
   $k = 12$
5. What is the measure of $m\angle BYD$, in degrees?
   $m\angle BYD = 68°$
6. What is the measure of $m\angle LDY$, in degrees?
   $m\angle LDY = 57°$

When introducing the problem, ask students what they know about drones.
Introducing this concept, help students understand and pay close attention to the key details that are presented within the context of the problem. Connect the instruction during this activity to what they did in the prior lesson so they can connect how these real-world problems can be solved using the same skills and knowledge.

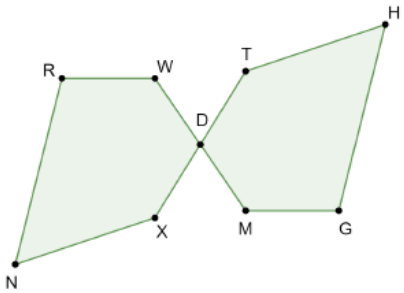
When reviewing these questions during instruction, consider using markers or highlighters to connect where the information needed came from in the original image.

5.10.3 Collaboration: Corresponding Parts in the Real World

Component Overview: *Students will work together to solve real-world problems using corresponding parts. For this activity, students should be allowed to work in pairs or small groups. Consider reviewing the context prior to releasing to students.*

5.10.3 Collaboration: Corresponding Parts in the Real World

Work with your partner to complete the following.



French formal gardens use geometric shapes that repeat throughout the garden. Dajon is a landscape designer. He is designing a garden in the tradition of a French formal garden. While at the client's home, he wrote down some measurements, in feet (ft), and then sketched polygons $RWDXN$ and $DTHGM$ to represent the garden. In the tradition of the congruence of polygons found in French formal gardens, Dajon also wrote $NRWDX \cong HGMDT$.

The garden's measurements are given in the table.

Garden Measurements	
$WD = (4x - 0.1)$ ft	$MD = 10.3$ ft
$RW = 12$ ft	$TH = (8x - 1.8)$ ft
$DX = (3x + 3.4)$ ft	$HG = (10x - 1.3)$ ft

1. Determine the perimeter of the garden. Show your work.

$WD = MD$

$4x - 0.1 = 10.3$

$x = 2.6$

$2(RW + WD + DX + NX + NR) =$ *perimeter of the garden*
$2(12 + 10.3 + 3(2.6) + 3.4 + 8(2.6) - 1.8 + 10(2.6) - 1.3) =$ *perimeter of the garden*
$2(12 + 10.3 + 7.8 + 3.4 + 20.8 - 1.8 + 26 - 1.3) =$ *perimeter of the garden*
$2(77.2) =$ *perimeter of the garden*
Perimeter of the garden is 154.4 *ft*

Prior to releasing for this activity, it may be beneficial to read and review the written context. This may assist struggling readers with a head start in thinking through the context when working with their partner.

For question 1, students may use either polygon when writing the equation of the perimeter. When reviewing answers, allow both to be shared and validated as being correct answers.

5.10.3 Collaboration: Corresponding Parts in the Real World (continued)

Once the garden has been installed, Dajon will check the angles of the polygons.

$m\angle HTD = (5k - 6)°$
$m\angle XNR = 58°$
$m\angle NRW = (4k + 4)°$
$m\angle DMG = (6k - 20)°$
$m\angle MGH = 104°$

2. Determine what measurement, in degrees, Dajon should use to check $m\angle XDW$. Show your work.

$$m\angle NRW = m\angle MGH$$

$$4k + 4 = 104$$
$$4k = 100$$
$$k = 25$$

$$m\angle NRW + m\angle RWD + m\angle XDW + m\angle DXN + m\angle XNR = 540°$$
$$(4(25) + 4) + (6(25) - 20) + m\angle XDW + (5(25) - 6) + 58 = 540°$$
$$104 + 130 + m\angle XDW + 119 + 58 = 540°$$
$$m\angle XDW + 411 = 540°$$
$$m\angle XDW = 129°$$

For question 2, students learned about interior angles sums in 8th grade Pre-Algebra. However, there may be students who took Algebra 1 instead of Pre-Algebra, potentially missing that content. Consider a quick review (no more than 2 minutes) of decomposing polygons into triangles to find the sum of the angles.

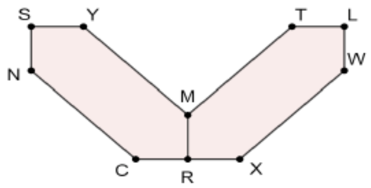
Facilitation questions to consider:
- *How do you know which measurement Dajon should use to check that angle?*
- *Why aren't other angles good options for checking that angle?*

5.10.4 Your Turn!

Component Overview: *Students will practice solving a real-world problem using corresponding parts. For this activity, allow students to work independently to determine their level of understanding of the instruction provided in the prior activities. After a given set of time, review the answers as a whole group.*

5.10.4 Your Turn!

1. Mireia is learning how to make stained glass in art class. She has created a design and labeled the vertices to help her keep track of the measurements. Her design is shown.



Mireia's design was created so that $RMYSNC \cong RMTLWX$, $\overline{CN} \cong \overline{MY}$, and $\overline{CR} \cong \overline{RM} \cong \overline{YS} \cong \overline{SN}$. Mireia needs to determine how much solder she needs. Solder is used along the perimeter, in inches (in.), of the design to set the glass in place.

Use the given measurements, in inches, to determine the amount of solder.

$WX = 9x - 2.3$

$YM = 6x + 1.3$

$MR = 3x - 0.6$

$$WX = YM$$
$$9x - 2.3 = 6x + 1.3$$
$$3x = 3.6$$
$$x = 1.2$$

$YM = TM = WX = NC = 6(1.2) + 1.3 = 8.5$

$MR = SY = SN = CR = XR = LW = TL = 3(1.2) - 0.6 = 3$

Perimeter $= YM + YS + SN + NC + CR + RX + XW + WL + LT + TM + MR$

Perimeter $= 8.5 + 3 + 3 + 8.5 + 3 + 3 + 3 + 8.5 + 3 + 3 + 8.5 + 3$
Perimeter $= 55$ *inches of solder*

The time allotted for this activity would be a great opportunity to pull small groups of students who may need additional support based on their performance in the prior activity and prior lesson.

Students may wonder how to handle the line segment MR. Consider monitoring students to ensure that they understand it as a shared side between the two pieces of stained glass and that solder would not need to be placed there twice.

5.10.5 Wrap-Up: Reflection

Component Overview: *Students will complete a self-reflection on their understanding of the different concepts of this unit. This activity should be done independently to get student-level individual reflection data.*

5.10.5 Wrap-Up: Reflection

1. Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentence by checking all that apply.
   Answers will vary.

I don't understand
- ◻ SSS
- ◻ SAS
- ◻ ASA
- ◻ AAS
- ◻ HL
- ◻ CPCTC

... YET.

- ◻ Naming congruent parts
- ◻ Paragraph proofs
- ◻ Two-column proofs
- ◻ Flowchart proofs
- ◻ Finding values in congruent polygons

do(does) NOT make sense ... YET.

I can explain
- ◻ SSS.
- ◻ SAS.
- ◻ ASA.
- ◻ AAS.
- ◻ HL.
- ◻ CPCTC.
- ◻ naming congruent parts.
- ◻ finding values in congruent polygons.

This activity provides students with a self-reflection on the concepts of the entire unit. Use their responses, coupled with their overall performance, to determine which students may need additional support before the end of unit assessment.